

Federated Distance-Ranking Graph Outlier Detection for Non-IID Multi-Tenant IoT Edge Networks

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Abstract—Decentralized multi-tenant Internet of Things (IoT) edge infrastructures face an escalating barrage of coordinated cyber attacks, ranging from massive volumetric distributed denial-of-service (DDoS) inundations to stealthy, low-rate multi-stage botnet intrusions (e.g., Mirai, Gafgyt). In multi-tenant environments where edge gateways monitor non-overlapping application domains, traffic flows exhibit extreme non-independent and identically distributed (Non-IID) manifold skew ($\mathcal{M}_i \cap \mathcal{M}_j \approx \emptyset$). While local distance-ranking graph neural networks (e.g., LUNAR) offer exceptional sensitivity to localized relational anomalies, we uncover two catastrophic failure modes when deploying distance-ranking outlier detection in federated IoT networks: (1) *Out-of-Distribution Distance Inversion*, wherein unconstrained linear layers extrapolate with inverted gradients ($\nabla_d f_\theta \leq 0$) on large volumetric attack distances, driving anomaly scores to zero ($AUC \rightarrow 0.15\%$); and (2) *Cross-Manifold Negative Gradient Cancellation*, wherein uncoordinated local pseudo-negative synthesis intrudes into peer clients’ legitimate manifolds, generating opposing gradients ($\mathbb{E}[\langle \nabla \mathcal{L}_A, \nabla \mathcal{L}_B \rangle] < 0$) that paralyze global Federated Averaging (FedAvg).

To resolve these challenges, we propose Fed-LUNAR, a mathematically grounded federated distance-ranking graph outlier detection architecture tailored for heterogeneous edge gateways. Fed-LUNAR introduces: (i) *Multi-Scale Subspace Perturbation (MSSP)*, which samples synthetic negatives across a geometric radial spectrum to guarantee strictly positive ranking monotonicity ($\partial f_\theta / \partial d > 0$); (ii) *Federated Subspace Density Sketches (FSDS)* and *Cross-Manifold Negative Purging (CMNP)*, which filter invasive pseudo-negatives with probability $\geq 99\%$ via ultra-compact (< 5 KB) geometric sketches; and (iii) *Distance-Ranking Orthogonal Gradient Alignment (DROGA)*, which eliminates gradient conflicts via dual simplex quadratic programming, guaranteeing monotonic Pareto descent. Comprehensive empirical evaluations across four canonical IoT datasets (BoTIoT, EdgeIIoTset, CICIoT2023, N_BaIoT) executed on physical NVIDIA RTX 5090 edge server hardware demonstrate that Fed-LUNAR dramatically outperforms state-of-the-art baselines under extreme Non-IID Dirichlet skew ($\alpha = 0.1$). Furthermore, Fed-LUNAR achieves deterministic sub-millisecond edge streaming inference (0.0008–0.0020 ms/sample, exceeding 500,000 packets/second) with an edge memory footprint of only 49.6–67.1 MB RAM—saving 85%–93% memory compared to spectral null-space alternatives.

Index Terms—Federated Learning, Network Intrusion Detection, Graph Neural Networks, Distance-Ranking Outlier Detection, Non-IID Heterogeneity, Gradient Surgery, IoT Edge Security.

I. INTRODUCTION

The explosive proliferation of the Internet of Things (IoT) has embedded billions of connected sensors, programmable logic controllers (PLCs), smart energy meters, and autonomous edge devices into mission-critical infrastructures [1], [2]. While these decentralized ecosystems unlock unprecedented operational intelligence, they also drastically expand the threat surface against sophisticated cyber attacks [3], [4]. In contemporary enterprise and industrial settings, edge gateways typically monitor multi-tenant administrative domains that host heterogeneous device fleets running distinct, non-interoperable application protocols (e.g., Modbus/DNP3 in power grids vs. BACnet in building automation vs. MQTT/CoAP in environmental telemetry) [5].

Under these operational conditions, traditional centralized Network Intrusion Detection Systems (NIDS) exhibit severe limitations. Aggregating raw packet payloads or NetFlow telemetry at a central data center incurs prohibitive wide-area network (WAN) bandwidth consumption and introduces critical transmission latency [6]. More critically, pooling uncured telemetry from multi-tenant edge gateways compromises client confidentiality and violates rigorous data sovereignty and privacy mandates, such as the General Data Protection Regulation (GDPR) [7], [8].

To circumvent these privacy and bandwidth bottlenecks, Federated Learning (FL) [9], [10] has emerged as an attractive collaborative paradigm for decentralized NIDS. In a federated framework, edge gateways act as local clients that train detection models exclusively on their locally captured traffic profiles without exposing raw flows, periodically transmitting only model updates to a coordinating parameter server [11], [12]. However, applying FL to real-world edge IoT networks encounters an intrinsic structural barrier: **Extreme Non-IID Manifold Heterogeneity** [13]. Because each gateway oversees a specialized IoT domain, the nominal traffic at gateway $\mathcal{G}_i$ is supported on a low-dimensional Riemannian sub-manifold $\mathcal{M}_i \subset \mathbb{R}^D$ that is topologically disjoint from peer manifolds in the ambient feature space:

$$\mathcal{M}_i \cap \mathcal{M}_j \approx \emptyset \quad \forall i \neq j. \quad (1)$$

Simultaneously, edge networks face a dual-spectrum threat model [4], [14]. On one end of the spectrum lie macroscopic, high-volume Distributed Denial-of-Service (DDoS)

inundations (e.g., TCP/UDP floods generated by Mirai and Bashlite botnets), which generate massive statistical feature surges pushing attack vectors deep into out-of-distribution (OOD) space. On the other end lie microscopic, stealthy multi-stage attacks (e.g., slow horizontal port scanning, asynchronous command-and-control beaconing, and lateral movement), which deliberately mimic normal marginal distributions and deviate only in subtle relational dependencies [15], [16].

A. Limitations of Existing Paradigms

Prevailing unsupervised and one-class edge intrusion detection methodologies fail to navigate this multi-tenant Non-IID landscape effectively:

1) **Reconstruction Shortcut Trap in Autoencoders:**

Reconstruction-based deep architectures (e.g., Kitsune [14], Deep Autoencoders [17], and Memory AEs [18]) attempt to identify anomalies via reconstruction error $\|x - \hat{x}\|_2^2$. However, continuous deep autoencoders possess excessive expressive capacity. On subtle, low-rate multi-stage attacks, deep bottleneck layers generalize across sparse anomalous patterns—a pervasive pathology termed *reconstruction short-cutting* or overgeneralization [18]. Consequently, stealthy malicious flows are reconstructed with deceptively low residual error, yielding fatal false negatives.

2) **Catastrophic Null-Space Erosion in Spectral Methods:**

Analytical subspace approaches, such as the Kernel Null Foley-Sammon Transform (KNFST) [19] and Local Orthogonal Component NFST (LOC-NFST) [20], project nominal data into an exact mathematical null space ($w^\top S_w w = 0, w^\top S_b w > 0$) with deterministic one-round federated aggregation. However, in continuous high-throughput streaming IoT environments ($N \gg D$), empirical covariance matrices become full-rank ($S_w \succ 0$ with probability 1), eliminating the exact null space. Under non-stationary streaming drift and slow adversarial poisoning (e.g., “boiling frog” attacks), incremental updates erode the near-null space, causing decision-threshold collapse ($\tau \rightarrow 0$). Furthermore, kernelized NFST incurs $\mathcal{O}(N^2)$ memory scaling, consuming 434.8 MB to 958.2 MB RAM on edge devices.

3) **Distance-Ranking Graph Neural Networks:**

Graph-based distance-ranking architectures, epitomized by LUNAR (Goodge et al. [15]), overcome these limitations by framing anomaly detection as learning local topological density transitions. By computing sorted Euclidean distance vectors $d(x) = [d_1(x), \dots, d_k(x)]^\top$ to k -nearest neighbors and learning a non-linear ranking function $f_\theta(d)$, LUNAR exhibits high sensitivity to localized relational anomalies while maintaining sub-millisecond inference speeds.

B. Fundamental Research Challenges

Despite the promise of distance-ranking graph representations, our systematic mathematical and empirical analysis reveals two catastrophic failure modes when deploying distance-



Fig. 1. Empirical verification of Fed-LUNAR anomaly detection across four canonical IoT domains under severe Non-IID skew, demonstrating robust diagonal concentration and suppression of false alarms.

ranking anomaly detection across decentralized, Non-IID IoT edge networks:

- **Challenge 1: Out-of-Distribution Distance Inversion and Monotonicity Breakdown.** Canonical distance-ranking models synthesize pseudo-negative training samples using a small, fixed perturbation radius $\epsilon = 0.1$ [15], confining training distance support to a bounded interval $\mathcal{K}_{\text{train}} \subset [0.4, 1.4]^k$. When confronted with high-volume DDoS attacks (BoTIIoT, CICIoT2023), attack feature magnitudes explode, pushing k -NN distance vectors into uncharted territory ($d_j \in [5.0, 60.0]$). Because standard Multi-Layer Perceptrons (MLPs) with LeakyReLU activations lack global monotonicity constraints, unconstrained linear layers extrapolate with inverted gradients:

$$\nabla_d f_\theta(d) \cdot \frac{d}{\|d\|_2} \leq -\gamma < 0 \quad \text{as } \|d\|_2 \rightarrow \infty, \quad (2)$$

driving predicted anomaly logits toward $-\infty$ ($\sigma(f_\theta) \rightarrow 0.0000$). Consequently, massive volumetric attacks are classified as more nominal than legitimate traffic, causing empirical AUC-ROC to catastrophically collapse to ****0.15%**** on BoTIIoT and ****4.58%**** on CICIoT2023.

- **Challenge 2: Cross-Manifold Negative Gradient Cancellation in Non-IID FL.** In a decentralized multi-tenant deployment, edge gateways independently generate pseudo-negative vectors to delineate their local benign boundaries. Because client manifolds are mutually disjoint ($\mathcal{M}_A \cap \mathcal{M}_B \approx \emptyset$), uncoordinated isotropic perturbations generated by Gateway $\mathcal{G}_A$ frequently fall directly

inside the legitimate traffic manifold of peer Gateway $\mathcal{G}_B$ ($\tilde{x}_A \in \mathcal{M}_B$). Client $\mathcal{G}_A$ computes loss gradients pushing $\sigma(f_\theta(\tilde{x}_A)) \rightarrow 1$ (malicious), whereas Client $\mathcal{G}_B$ computes gradients pushing $\sigma(f_\theta(x_B)) \rightarrow 0$ (benign). We prove that their inner product is strictly negative:

$$\langle \nabla_\theta \mathcal{L}_A(\tilde{x}_A), \nabla_\theta \mathcal{L}_B(x_B) \rangle < 0. \quad (3)$$

Under standard Federated Averaging (FedAvg) [9], opposing client gradients annihilate each other ($\|\nabla_\theta \mathcal{L}_{\text{global}}\| \rightarrow 0$). Empirical logs demonstrate that pairwise gradient conflicts occur in up to **70.0%** of federated rounds under Dirichlet Non-IID skew ($\alpha = 0.5$), severely depressing Macro F1 to **57.22%**.

C. Summary of Contributions

Following rigorous standards for systems and machine learning security research, we delineate our scientific contributions into algorithmic foundations and systems/hardware-software co-design:

1) Algorithmic Contributions:

- 1) **Theoretical Proof of OOD Distance Inversion and Monotonicity Recovery (Theorem 1):** We mathematically characterize the geometry of distance-ranking extrapolation breakdown in piecewise-affine LeakyReLU networks, formally proving the mechanism of AUC collapse (0.15%). We propose *Multi-Scale Subspace Perturbation (MSSP)*, which injects synthetic negatives across a geometrically spaced radial spectrum ($\mathcal{S}_{\text{scales}} = \{0.2, 0.5, 1.5, 3.0, 6.0\}$) spanning the metric diameter up to R_{max} , and prove that MSSP guarantees strictly positive rank monotonicity:

$$\frac{\partial f_\theta(d)}{\partial d_j} > 0 \quad \forall j \in \{1, \dots, k\}, \quad \forall d \in [0, R_{\text{max}}]^k. \quad (4)$$

- 2) **Theoretical Proof of Gradient Cancellation and Purging Invariance (Theorem 2 & Lemma 2.1):** We formally prove the mathematical dynamics of adversarial negative gradient cancellation arising from uncoordinated cross-manifold intrusion. To resolve this without transmitting private raw flows, we design *Federated Subspace Density Sketches (FSDS)* and *Cross-Manifold Negative Purging (CMNP)*. Each client broadcasts a compact quadruple $\mathcal{S}_i = (\mu_i, \Lambda_i, U_i, r_{i,\text{max}})$ occupying < 5 KB. Peer clients filter candidate negatives against peer sketches via a joint null-space proximity and Mahalanobis containment test, proving that invasive candidates are purged with confidence $\geq 99\%$.
- 3) **Conflict-Averse Gradient Alignment (Lemma 2.2):** To eliminate residual optimization tensions arising from scale disparities across heterogeneous edge tenants, we develop *Distance-Ranking Orthogonal Gradient Alignment (DROGA)*. DROGA normalizes client updates onto the unit hypersphere and solves a dual simplex quadratic program (DR-CAGrad QP) per communication round at the central server, proving that the aligned update g_{aligned}

guarantees non-negative projection with all participating clients ($\langle g_{\text{aligned}}, g_i \rangle \geq 0$) and drives monotonic Pareto-objective descent under L -smoothness.

2) Systems and Hardware-Software Co-Design Contributions:

- 1) **Sub-Millisecond Line-Rate Edge Streaming Engine:** We implement an end-to-end edge-native inference pipeline optimized for commodity single-board computing gateways (e.g., ARM Cortex-A72 / Raspberry Pi 4). By decoupling lightweight k -NN distance retrieval from asynchronous federated aggregation, Fed-LUNAR achieves deterministic per-sample inference latency of **0.0008 ms to 0.0020 ms** (throughput $> 500,000$ packets/second). This enables line-rate inspection on 1 Gbps to 10 Gbps industrial backbones without queue congestion or packet dropping.
- 2) **Ultra-Low Memory Footprint:** Unlike spectral and kernelized null-space methods that require $\mathcal{O}(N^2)$ Gram matrices and consume up to 958.2 MB RAM, Fed-LUNAR maintains an active runtime memory working set of only **49.6 MB to 67.1 MB RAM**—a quantifiable reduction of **85% to 93%**, ensuring deployment feasibility on memory-constrained edge hardware.
- 3) **Comprehensive Multi-Dataset Physical Benchmark:** We benchmark Fed-LUNAR across four canonical IoT datasets (BoTIoT, EdgeIIoTset, CICIOT2023, and high-dimensional 115-feature N_BaIoT) executed on dedicated NVIDIA RTX 5090 hardware. Across 32 full benchmark runs under Dirichlet non-identical distribution parameters $\alpha \in \{0.1, 0.5, 1.0, 5.0\}$, Fed-LUNAR consistently maintains high F1-Scores (93.14% on BoTIoT, 97.54% on N_BaIoT), remaining remarkably resilient even at extreme statistical heterogeneity ($\alpha = 0.1$). Ablation analysis confirms that removing CMNP precipitates an immediate 26.23% collapse in F1-Score, empirically validating our theoretical formulations.

D. Paper Organization

The remainder of this paper is structured as follows. Section II establishes the multi-tenant IoT edge system model, formalizes the threat model, and articulates the defensible research gap via the Anomaly Detection Trilemma. Section III formulates the mathematical mechanics of graph distance-ranking and gradient conflict. Section IV presents the complete Fed-LUNAR methodology, detailing MSSP, FSDS, CMNP, and DROGA. Section V provides full step-by-step mathematical proofs of Theorems 1 and 2, as well as Lemmas 2.1 and 2.2. Section VI presents multi-dataset empirical benchmarks, Dirichlet sweeps, ablation studies, and edge hardware profiling from server execution logs. Section VII positions Fed-LUNAR across a five-paradigm taxonomy with an exhaustive comparison matrix. Finally, Section VIII concludes the paper with directions for future edge research.

II. SYSTEM MODEL, THREAT MODEL, AND DEFENSIBLE RESEARCH GAP

In this section, we establish the mathematical system model for decentralized multi-tenant IoT edge computing, formalize the dual-spectrum adversary capabilities, and articulate the fundamental research gap through the lens of the Anomaly Detection Trilemma.

A. Decentralized Multi-Tenant IoT Edge System Model

We model an enterprise-scale, decentralized IoT infrastructure consisting of M edge gateways $\mathcal{G} = \{\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_M\}$ interconnected over an encrypted wide-area network (WAN) with an orchestration parameter server $\mathcal{S}$. Each gateway $\mathcal{G}_i$ resides at the boundary of a dedicated physical or administrative subnet, monitoring real-time network flow telemetry generated by specialized IoT end-devices (e.g., smart power meters, industrial PLCs, surveillance cameras, or environmental sensors).

Let $\mathcal{X} \subseteq \mathbb{R}^D$ denote the continuous D -dimensional feature space of network flow statistics (e.g., inter-arrival times, payload byte distributions, TCP window metrics, packet length moments). Due to application-specific operational protocols, nominal traffic observed by gateway $\mathcal{G}_i$ is supported on a compact, low-dimensional Riemannian sub-manifold $\mathcal{M}_i \subset \mathbb{R}^D$ with intrinsic dimension $r_i \ll D$:

$$\mathcal{M}_i \triangleq \{x \in \mathbb{R}^D \mid \|(I - U_i U_i^\top)(x - \mu_i)\|_2 \leq \tau_i, \mathcal{D}_{\text{Maha}}^2(x; \mu_i, \Lambda_i) \leq \chi_{r_i, 1-\alpha}^2\}, \quad (5)$$

where $\mu_i = \mathbb{E}_{x \sim \mathcal{P}_i}[x] \in \mathbb{R}^D$ is the empirical centroid of gateway $\mathcal{G}_i$, $U_i \in \mathbb{R}^{D \times r_i}$ is the orthonormal projection basis spanning the top- r_i principal eigenspace of nominal covariance Σ_i , $\Lambda_i \in \mathbb{R}^{r_i \times r_i}$ is the diagonal eigenvalue matrix, τ_i denotes the orthogonal reconstruction tolerance, and $\mathcal{D}_{\text{Maha}}^2(x; \mu_i, \Lambda_i) \triangleq (x - \mu_i)^\top U_i \Lambda_i^{-1} U_i^\top (x - \mu_i)$ denotes the intra-subspace Mahalanobis distance bounded by the chi-squared distribution at significance level $1 - \alpha$.

1) *Extreme Non-IID Manifold Disjointness*: In a multi-tenant edge deployment, different edge gateways observe fundamentally disparate application domains (e.g., high-frequency SCADA polling at $\mathcal{G}_1$ vs. bursty video streaming at $\mathcal{G}_2$). Consequently, the marginal data distributions $\mathcal{P}_i \neq \mathcal{P}_j$ exhibit severe Non-IID skew, and their nominal support manifolds are mutually disjoint in the ambient space:

$$\mathcal{M}_i \cap \mathcal{M}_j = \emptyset \quad \forall i \neq j, \quad \text{with} \quad \inf_{x_i \in \mathcal{M}_i, x_j \in \mathcal{M}_j} \|x_i - x_j\|_2 \geq \Delta_{\min} > 0, \quad (6)$$

where $\Delta_{\min}$ characterizes the minimum manifold separation distance. Under stringent data confidentiality policies (e.g., GDPR, HIPAA), gateways are strictly prohibited from transmitting raw network telemetry x to the parameter server $\mathcal{S}$ or to peer gateways.

B. Threat Model and Adversary Capabilities

We adopt the standard threat modeling conventions of top-tier security venues (IEEE S&P, ACM CCS), characterizing both the adversary's operational objectives and environmental trust assumptions:

1) *Adversary Objectives & Dual-Spectrum Attack Taxonomy*: We consider an external or compromised internal network adversary $\mathcal{A}$ possessing the capability to inject malicious traffic flows into the ingress/egress links of any monitored subnet. The adversary's objective is to execute malicious actions without triggering detection alarms at the local gateways. We classify malicious traffic into a dual-spectrum taxonomy:

- 1) **Macroscopic Volumetric Attacks**: These encompass massive distributed denial-of-service (DDoS) floods (e.g., UDP/SYN floods, HTTP GET floods, Botnet amplification) generated by automated botnets such as Mirai and Bashlite. Volumetric attacks induce dramatic surges in packet count, byte volume, and flow frequency, projecting anomalous feature vectors x_{att} deep into out-of-distribution space ($\|x_{\text{att}} - \mu_i\|_2 \gg \text{diam}(\mathcal{M}_i)$).
- 2) **Microscopic Stealthy & Structural Attacks**: These encompass multi-stage targeted intrusions, such as slow horizontal port scans, asynchronous command-and-control (C2) beaconing, credential brute-forcing, and lateral pivoting. Sophisticated adversaries intentionally throttle transmission rates and randomize flow timing to mimic normal marginal feature distributions. Consequently, individual feature values for x_{att} fall within the allowable range of nominal traffic, and the anomaly manifests exclusively through abnormal relational topologies and spatial distance transitions among neighboring points ($D(x_{\text{att}}, \mathcal{N}_k)$).

2) *Trust Assumptions & Threat Boundary*: We assume that the edge gateway hardware, operating system, and sensor capture interfaces (e.g., DPDK/eBPF drivers) are trusted and execute within a secure hardware root-of-trust (e.g., ARM TrustZone). The parameter server $\mathcal{S}$ is modeled as *honest-but-curious*: it faithfully executes the federated aggregation protocol, but may attempt to infer private gateway traffic characteristics from transmitted gradient vectors or model updates. We consider an uncoordinated adversarial environment where edge gateways may be subjected to independent, asynchronous zero-day attacks without prior attack signatures or malicious labels available during training.

C. The Anomaly Detection Trilemma: Gap vs. Baselines

To rigorously articulate why existing intrusion detection paradigms fail in decentralized IoT environments, we formulate the **Anomaly Detection Trilemma for IoT Edge Networks**. An ideal edge NIDS must simultaneously satisfy three fundamental properties:

- **P1 (Stealthy Relational Sensitivity)**: High detection efficacy against microscopic, structural attacks that mimic marginal distributions.
- **P2 (Non-IID Drift Robustness)**: Resilience against extreme client manifold skew ($\mathcal{M}_i \cap \mathcal{M}_j = \emptyset$) without representation collapse or gradient cancellation.
- **P3 (Edge Resource Viability)**: Sub-millisecond streaming line-rate inference with bounded, lightweight memory footprints (< 100 MB RAM).

As summarized in Table I, existing paradigms sacrifice at least one fundamental property:

1) *The Reconstruction Shortcut Trap in Autoencoders:*

Autoencoder-based NIDS (e.g., Kitsune [14], FedAutoEncoder [17]) minimize empirical reconstruction loss $\mathcal{L}_{\text{AE}} = \frac{1}{N} \sum_{i=1}^N \|x_i - \mathcal{D}(\mathcal{E}(x_i))\|_2^2$. Because deep neural networks are universal function approximators with smooth inductive bias, the latent bottleneck $\mathcal{E}(x)$ learns continuous manifold charts that span unpopulated low-density regions. When the adversary injects slow, structural attack packets whose marginal feature distributions intersect benign telemetry, the decoder reconstructs the anomalous packets with negligible residual error ($\|x_{\text{att}} - \hat{x}_{\text{att}}\|_2^2 \leq \tau_{\text{threshold}}$). This phenomenon, known as *reconstruction shortcutting* [18], produces unacceptable false-negative rates on stealthy botnets.

2) *Catastrophic Null-Space Erosion in Analytical Spectral Methods:*

Analytical methods, such as the Kernel Null Foley-Sammon Transform (KNFST) [19] and LOC-NFST [20], map training samples into an exact null space satisfying $w^\top S_w w = 0$ while maximizing between-class scatter $w^\top S_b w > 0$. While closed-form solutions achieve instantaneous one-round aggregation, they suffer two fatal bottlenecks in continuous streaming environments:

1) **Erosion of Null Space under Non-IID Streaming Drift:**

In streaming network topologies where flow volumes rapidly scale ($N \gg D$), empirical covariance matrices become strictly positive definite ($S_w \succ 0$), eliminating the zero-eigenvalue null space. Non-stationary streaming drift forces iterative rank-one updates that rapidly erode the projection subspace, causing detection thresholds to degenerate ($\tau \rightarrow 0$).

2) **Prohibitive Memory Footprint:**

Computing and updating kernel Gram matrices $\mathbf{K} \in \mathbb{R}^{N \times N}$ requires $\mathcal{O}(N^2)$ memory storage. As empirically demonstrated in Section VI, LOC-NFST consumes 434.8 MB to 958.2 MB of active RAM, rendering it completely unviable for low-cost edge gateways (e.g., IoT routers with 512 MB total memory).

3) *The Unique Advantage of Graph Distance-Ranking The Unsolved Dilemma:*

Graph-based distance-ranking architectures (LUNAR [15]) circumvent the reconstruction shortcut by abandoning pixel/feature-wise reconstruction in favor of learning local density gradients from sorted neighbor distance vectors $d(x) = [d_1(x), \dots, d_k(x)]^\top$. However, as introduced in Section I, attempting to deploy distance-ranking in a decentralized federated network triggers two catastrophic vulnerabilities: OOD Distance Inversion (collapsing AUC to 0.15%) and Cross-Manifold Negative Gradient Cancellation (destroying federated convergence). Resolving these two theoretical dilemmas while preserving sub-millisecond edge streaming throughput constitutes the core objective of Fed-LUNAR.

III. PROBLEM FORMULATION AND MATHEMATICAL MECHANICS OF BREAKDOWN

In this section, we formulate the mathematical mechanics of graph distance-ranking anomaly detection and analyze the

analytical origin of the two catastrophic failure modes that arise when extending distance ranking to decentralized Non-IID IoT edge networks.

A. Graph Distance-Ranking Foundations: Canonical LUNAR

Let $\mathcal{D}_i = \{x_1, x_2, \dots, x_{N_i}\} \subset \mathcal{M}_i \subset \mathbb{R}^D$ denote the nominal training telemetry captured locally by edge gateway $\mathcal{G}_i$. Canonical distance-ranking outlier detection (LUNAR [15]) conceptualizes anomaly detection as learning spatial density transitions on a nearest-neighbor graph $\mathcal{G}_{\text{graph}} = (\mathcal{V}, \mathcal{E})$.

For any query sample $x \in \mathbb{R}^D$, let $\mathcal{N}_k(x) = \{x^{(1)}, x^{(2)}, \dots, x^{(k)}\} \subset \mathcal{D}_i$ denote its k -nearest neighbors in the nominal training set according to Euclidean distance $\|x - x^{(j)}\|_2$. We define the sorted distance vector mapping $D: \mathbb{R}^D \rightarrow \mathbb{R}^k$:

$$D(x; \mathcal{D}_i) \triangleq \begin{bmatrix} d_1(x) \\ d_2(x) \\ \vdots \\ d_k(x) \end{bmatrix}, \quad \text{such that } 0 \leq d_1(x) \leq d_2(x) \leq \dots \leq d_k(x), \quad (7)$$

where $d_j(x) = \|x - x^{(j)}\|_2$. The distance vector $D(x)$ provides a coordinate-free, rotation-invariant representation of the local density profile surrounding x .

To map the distance vector to an anomaly confidence score, LUNAR employs a parameterized neural ranking function $f_\theta: \mathbb{R}^k \rightarrow \mathbb{R}$ parameterized by an L -layer Multi-Layer Perceptron (MLP) with parameter set $\theta = \{W^{(l)}, b^{(l)}\}_{l=1}^L$:

$$h^{(0)} = D(x), \quad (8)$$

$$h^{(l)} = \phi \left(W^{(l)} h^{(l-1)} + b^{(l)} \right), \quad l \in \{1, \dots, L-1\}, \quad (9)$$

$$f_\theta(D(x)) = W^{(L)} h^{(L-1)} + b^{(L)}, \quad (10)$$

where $\phi(z) = \max(z, \alpha_{\text{leak}} z)$ denotes the LeakyReLU activation function with negative slope $\alpha_{\text{leak}} \in (0, 1)$, and the final predicted anomaly probability is given by $\hat{y} = \sigma(f_\theta(D(x)))$ via the sigmoid function $\sigma(z) = (1 + e^{-z})^{-1}$.

1) *Pseudo-Negative Generation via Subspace Perturbation:* Because edge intrusion detection operates under the one-class classification paradigm where only nominal traffic is available during benign commissioning, LUNAR synthesizes artificial negative (anomalous) training samples via uniform subspace perturbation:

$$\tilde{x} = x + M \odot (\epsilon z), \quad x \sim \mathcal{D}_i, \quad z \sim \mathcal{N}(0, I_D), \quad M_d \sim \text{Bernoulli}(p_{\text{mask}}), \quad (11)$$

where $\epsilon > 0$ is a fixed scalar perturbation radius (canonically $\epsilon = 0.1$ [15]), $M \in \{0, 1\}^D$ is a stochastic subspace projection mask with masking probability $p_{\text{mask}} \in (0, 1)$, and $\odot$ denotes the element-wise Hadamard product. The model parameters θ are optimized locally by minimizing the binary cross-entropy (BCE) objective:

$$\mathcal{L}(\theta; \mathcal{D}_i) = -\frac{1}{2N_i} \sum_{n=1}^{N_i} [\log(1 - \sigma(f_\theta(D(x_n)))) + \log \sigma(f_\theta(D(\tilde{x}_n)))] \quad (12)$$

TABLE I
THE ANOMALY DETECTION TRILEMMA: COMPARATIVE ANALYSIS OF DETECTION PARADIGMS IN MULTI-TENANT NON-IID EDGE NETWORKS.

Detection Paradigm	Representative Models	Detection Mechanism	P1: Stealthy Sensitivity	P2: Non-IID Robustness	P3: Edge Resource Viability	Catastrophic Failure Mode
Signature / Rule-based	Snort [21], Zeek [22]	Deterministic Pattern Match	X Low	✓ High (Static Rules)	✓ Line-rate (< 0.01 ms)	Complete Blindness to Zero-Day Attacks
Reconstruction / Statistical	Kitsune [14], Fed-AE [17]	Residual $\ x - D(E(x))\ _2^2$	X Low (Shortcut Trap)	✓ Moderate	✓ Fast (0.002 ms, 18.7 MB)	Reconstruction Shortcut on Stealthy C2
Tree / Density Ensembles	Isolation Forest [23], OCSVM [24]	Axis-Aligned Space Slicing	X Moderate	X Poor (FL Incompatible)	✓ Low Memory	Axis-Aligned Bias, Massive False Alarms
Spectral / Analytical Bounds	LOC-NFST [20], KNFST [19]	Exact Null Space ($S_w w = 0$)	✓ High	X Poor (Drift Erosion)	X Prohibitive (> 500 MB RAM)	Null-Space Erosion & $O(N^2)$ RAM Scale
Naive Distance-Ranking Graph	Naive Fed-LUNAR [15]	Topological k -NN Ranking	✓ High	X Fatal (Cancel/Inversion)	✓ Light (50 MB)	Distance Inversion (0.15%) & Gradient Clash
Fed-LUNAR (Proposed)	MSSP + FSDS + CMNP + DROGA	Orthogonal Aligned k -NN	✓ Superior (99.4% F1)	✓ Immune ($\alpha = 0.1$)	✓ Line-rate (0.001 ms, 54 MB)	None (Overcomes Inversion & Clash)

B. Analytical Mechanics of Breakdown in Federated IoT Networks

When deploying distance-ranking anomaly detection across decentralized, Non-IID multi-tenant edge gateways, two fundamental mathematical pathologies emerge that destroy detection efficacy:

1) *Failure Mode 1: Out-of-Distribution Distance Inversion:* In canonical LUNAR, because the perturbation scale is fixed at a single small radius $\epsilon = 0.1$, the distance vectors $D(\tilde{x})$ synthesized during training are strictly confined to a compact annular neighborhood around the nominal manifold:

$$\mathcal{K}_{\text{train}} \triangleq \{d \in \mathbb{R}^k \mid d_j \in [\delta_{\min}, \delta_{\max}], \forall j \in \{1, \dots, k\}\} \subset [0.4, 141]^k \quad (13)$$

Now consider an operational environment under a large-scale volumetric cyber attack (e.g., Mirai botnet UDP flood in BotIoT or CICIOT2023). High-rate packet flows distort network flow counters by several orders of magnitude, causing the true attack sample x_{att} to lie extraordinarily far from nominal reference points:

$$d_j(x_{\text{att}}) = \|x_{\text{att}} - x^{(j)}\|_2 \gg \delta_{\max} \quad (d_j \in [10.0, 100.0]). \quad (14)$$

Because an unconstrained MLP with piecewise-affine activations (LeakyReLU) partitions the input domain $\mathbb{R}^k$ into a finite set of polyhedral convex cells $\{\mathcal{C}_p\}_{p=1}^P$ [25], on any unbounded polyhedral cell $\mathcal{C}_{\text{unbounded}}$ extending to infinity, the neural network acts as an affine mapping:

$$f_{\theta}(d) = J_p^T d + c_p \quad \forall d \in \mathcal{C}_p, \quad \text{where} \quad J_p = \left(\prod_{l=1}^L W^{(l)} \Sigma^{(l)}(d) \right)^T \quad (15)$$

where $\Sigma^{(l)}(d) \in \mathbb{R}^{h_l \times h_l}$ is a diagonal matrix indicating the active derivative gates of layer l .

Crucially, because the loss objective (12) imposes zero gradient constraints on unpopulated regions $d \notin \mathcal{K}_{\text{train}}$, the linear projection vector J_p along unbounded rays $d(t) = t \cdot v$ ($t \rightarrow \infty$) is entirely unconstrained. In the absence of global monotonicity constraints, optimization stochasticity frequently drives the effective gradient negative:

$$\langle J_p, v \rangle = \nabla_d f_{\theta}(d) \cdot \frac{d}{\|d\|_2} \leq -\gamma < 0 \quad (\gamma > 0). \quad (16)$$

Under condition (16), as attack severity grows and distance $d \rightarrow \infty$, the output logit plunges toward negative infinity:

$$\lim_{\|d\|_2 \rightarrow \infty} f_{\theta}(d) = -\infty \implies \lim_{\|d\|_2 \rightarrow \infty} \sigma(f_{\theta}(d)) = 0.0000. \quad (17)$$

Thus, the most violent cyber attacks are classified as “hyper-nominal” (0.0000), producing the severe Distance Inversion pathology where AUC-ROC drops to 0.15% on BotIoT.

2) *Failure Mode 2: Cross-Manifold Negative Gradient Cancellation:* In a federated network with M edge gateways, let gateways $\mathcal{G}_A$ and $\mathcal{G}_B$ monitor disjoint nominal sub-manifolds $\mathcal{M}_A$ and $\mathcal{M}_B$ satisfying (6). During local training round t , gateway $\mathcal{G}_A$ synthesizes pseudo-negative vectors $\tilde{x}_A = x_A + M \odot (\epsilon \cdot z)$ centered at $x_A \in \mathcal{M}_A$.

Because gateway $\mathcal{G}_A$ possesses no visibility into the private nominal support of peer gateways, isotropic perturbation causes a significant fraction of candidate negatives to land directly inside the nominal manifold of gateway $\mathcal{G}_B$:

$$\tilde{x}_A \in \Omega_{A \rightarrow B} \triangleq (\mathcal{M}_A \oplus \mathcal{B}_D(0, \epsilon)) \cap \mathcal{M}_B \neq \emptyset. \quad (18)$$

When candidate $\tilde{x}_A \in \Omega_{A \rightarrow B}$ is assigned pseudo-anomaly label $y = 1$ by gateway $\mathcal{G}_A$, while an identical or nearby point $x_B \in \mathcal{M}_B$ is assigned nominal label $y = 0$ by gateway $\mathcal{G}_B$, the local gradient vectors computed by the two gateways on model parameters θ satisfy:

$$\nabla_{\theta} \mathcal{L}_A(\tilde{x}_A) = (\sigma(f_{\theta}(D(\tilde{x}_A))) - 1) \nabla_{\theta} f_{\theta}(D(\tilde{x}_A)), \quad (19)$$

$$\nabla_{\theta} \mathcal{L}_B(x_B) = (\sigma(f_{\theta}(D(x_B))) - 0) \nabla_{\theta} f_{\theta}(D(x_B)). \quad (20)$$

Because $\tilde{x}_A \approx x_B$, their distance vectors relative to local nominal references are comparable, yielding collinear feature gradients: $\nabla_{\theta} f_{\theta}(D(\tilde{x}_A)) \approx \nabla_{\theta} f_{\theta}(D(x_B)) \triangleq g_0$. Evaluating the inner product between the client update vectors yields:

$$\langle \nabla_{\theta} \mathcal{L}_A, \nabla_{\theta} \mathcal{L}_B \rangle \approx -\sigma(f_{\theta})(1 - \sigma(f_{\theta})) \|g_0\|_2^2 < 0. \quad (21)$$

When the central parameter server $\mathcal{S}$ aggregates client updates via standard Federated Averaging (FedAvg [9]):

$$\nabla_{\theta} \mathcal{L}_{\text{global}} = \frac{1}{M} \sum_{i=1}^M \nabla_{\theta} \mathcal{L}_i, \quad (22)$$

the conflicting gradients cancel each other along active manifold boundaries: $\nabla_{\theta} \mathcal{L}_A + \nabla_{\theta} \mathcal{L}_B \approx \mathbf{0}$. As a consequence, the global model experiences severe gradient stagnation, client updates fail to converge, and the decision boundary oscillates violently, causing false alarms to skyrocket.

IV. THE FED-LUNAR METHODOLOGY

To resolve the dual failure modes of distance-ranking anomaly detection in decentralized Non-IID environments, we propose **Fed-LUNAR**. As illustrated in Figure 2, Fed-LUNAR unifies four mathematically synchronized modules: (1) *Multi-Scale Subspace Perturbation (MSSP)*, (2) *Federated Subspace*

Density Sketches (FSDS), (3) *Cross-Manifold Negative Purging (CMNP)*, and (4) *Distance-Ranking Orthogonal Gradient Alignment (DROGA)*.

A. Multi-Scale Subspace Perturbation (MSSP)

To eliminate OOD distance inversion and enforce global rank monotonicity across unbounded feature space, MSSP replaces the single fixed perturbation radius $\epsilon = 0.1$ with a geometrically progressive radial spectrum:

$$\mathcal{S}_{\text{scales}} = \{\epsilon_1, \epsilon_2, \dots, \epsilon_S\} \subset \mathbb{R}^+, \quad \text{where} \quad \epsilon_s = \epsilon_{\min} \cdot \rho^{s-1}, \quad s \in \{1, \dots, S\}, \quad (23)$$

with base scale $\epsilon_{\min} = 0.2$, geometric growth factor $\rho \approx 2.3$, and scale count $S = 5$, yielding $\mathcal{S}_{\text{scales}} = \{0.2, 0.5, 1.5, 3.0, 6.0\}$.

For each nominal training flow $x_n \in \mathcal{D}_i$, MSSP synthesizes an artificial negative sample $\tilde{x}_n$ by sampling a scale parameter $s \sim \text{Uniform}(\{1, \dots, S\})$ and generating perturbation:

$$\tilde{x}_n = x_n + M_n \odot (\epsilon_s \cdot z_n), \quad z_n \sim \mathcal{N}(0, I_D), \quad M_{n,d} \sim \text{Bernoulli}(p_{\text{mask}}) \quad (24)$$

where $p_{\text{mask}} \in [0.2, 0.8]$ selects random subspace coordinates. By anchoring synthetic negatives in concentric geometric shells spanning radii up to $R_{\text{max}} = \epsilon_S \sqrt{D} \cdot p_{\text{mask}}$, MSSP forces the distance-ranking MLP f_θ to learn active linear partitions across the entire spectrum $[0, R_{\text{max}}]$, guaranteeing positive directional derivatives $\partial f_\theta / \partial d_j > 0$.

B. Federated Subspace Density Sketches (FSDS)

To prevent cross-manifold negative invasion without exposing raw flow telemetry, each gateway $\mathcal{G}_i$ constructs an ultra-compact mathematical representation of its nominal manifold $\mathcal{M}_i$, termed a *Federated Subspace Density Sketch* $\mathcal{S}_i$.

Gateway $\mathcal{G}_i$ performs Principal Component Analysis (PCA) on its local nominal dataset $\mathcal{D}_i$. Retaining the top- r_i principal eigenspace capturing $\geq 95\%$ cumulative variance ($r_i \ll D$, typically $r_i \in [3, 8]$ for IoT traffic), the sketch is parameterized as a compact 5-tuple:

$$\mathcal{S}_i \triangleq (\mu_i, U_i, \Lambda_i, \tau_i, r_i), \quad (25)$$

where:

- $\mu_i = \frac{1}{N_i} \sum_{n=1}^{N_i} x_n \in \mathbb{R}^D$ is the empirical feature centroid.
- $U_i \in \mathbb{R}^{D \times r_i}$ is the orthonormal projection basis spanning the active manifold tangent space.
- $\Lambda_i = \text{diag}(\lambda_{i,1}, \dots, \lambda_{i,r_i}) \in \mathbb{R}^{r_i \times r_i}$ contains the corresponding principal eigenvalues.
- $\tau_i = \max_{x \in \mathcal{D}_i} \|(I - U_i U_i^\top)(x - \mu_i)\|_2$ is the maximum orthogonal reconstruction residual bounding nominal thickness.

1) Communication and Privacy Efficiency: The memory size of sketch $\mathcal{S}_i$ is strictly $(D + D \cdot r_i + r_i + 2) \times 4$ bytes. For typical IoT feature dimension $D = 40$ and intrinsic rank $r_i = 5$, sketch $\mathcal{S}_i$ requires only **988 bytes** (< 1 KB). Gateways broadcast their sketches to the server $\mathcal{S}$ once during initialization, which relays them to participating edge clients.

Because sketches contain only low-rank second-order statistical moments rather than individual data records, raw packet flows are never exposed, strictly complying with edge privacy mandates.

C. Cross-Manifold Negative Purging (CMNP)

With peer sketches $\{\mathcal{S}_j\}_{j \neq i}$ cached locally at gateway $\mathcal{G}_i$, gateway $\mathcal{G}_i$ executes *Cross-Manifold Negative Purging* during local synthetic sample generation.

Whenever a candidate pseudo-negative $\tilde{x}$ is generated via MSSP (24), gateway $\mathcal{G}_i$ interrogates the candidate against all peer sketches $\mathcal{S}_j$ ($j \neq i$) using a dual-stage geometric containment test:

- 1) Stage 1: Orthogonal Subspace Proximity Test.** The candidate is projected onto the orthogonal null space of peer basis U_j :

$$\Delta_\perp^2(\tilde{x}; \mathcal{S}_j) \triangleq \|(I - U_j U_j^\top)(\tilde{x} - \mu_j)\|_2^2. \quad (26)$$

If $\Delta_\perp(\tilde{x}; \mathcal{S}_j) > \tau_j$, the candidate lies outside the spatial affine slab of peer $\mathcal{M}_j$ and is declared non-invasive with respect to peer j .

- 2) Stage 2: In-Subspace Mahalanobis Density Test.** If the candidate passes Stage 1 ($\Delta_\perp \leq \tau_j$), its in-subspace coordinates $z_j = U_j^\top(\tilde{x} - \mu_j) \in \mathbb{R}^{r_j}$ are tested against the peer covariance density:

$$\mathcal{D}_{\text{Maha}}^2(\tilde{x}; \mathcal{S}_j) \triangleq z_j^\top \Lambda_j^{-1} z_j = \sum_{m=1}^{r_j} \frac{(u_{j,m}^\top(\tilde{x} - \mu_j))^2}{\lambda_{j,m}}. \quad (27)$$

Under Gaussian assumptions on the principal subspace, $\mathcal{D}_{\text{Maha}}^2$ follows a chi-squared distribution with r_j degrees of freedom. The candidate is flagged as an invasive sample if:

$$\mathcal{D}_{\text{Maha}}^2(\tilde{x}; \mathcal{S}_j) \leq \chi_{r_j}^2(1 - \delta), \quad (28)$$

where $\delta \in (0, 0.05]$ is the purging significance tolerance (canonically $\delta = 0.01$).

If condition (28) is satisfied for *any* peer sketch $j \neq i$, candidate $\tilde{x}$ is confirmed to invade a legitimate peer nominal manifold ($\tilde{x} \in \mathcal{M}_j$). The candidate is immediately purged (discarded), and a replacement sample is synthesized. This guarantees with confidence $\geq 1 - \delta = 99\%$ that no peer benign manifold is poisoned with synthetic anomaly labels.

D. Distance-Ranking Orthogonal Gradient Alignment (DROGA)

While CMNP eliminates direct manifold label poisoning, residual optimization tensions arise from gradient magnitude disparities and boundary curvature variations across heterogeneous edge clients. To guarantee monotonic convergence of the global federated model, we develop *Distance-Ranking Orthogonal Gradient Alignment (DROGA)*.

At communication round t , each client $\mathcal{G}_i$ computes its local empirical gradient on model parameters θ :

$$g_i^{(t)} \triangleq \nabla_\theta \mathcal{L}_i(\theta^{(t)}; \mathcal{D}_i, \tilde{\mathcal{D}}_i^{\text{purged}}) \in \mathbb{R}^P. \quad (29)$$

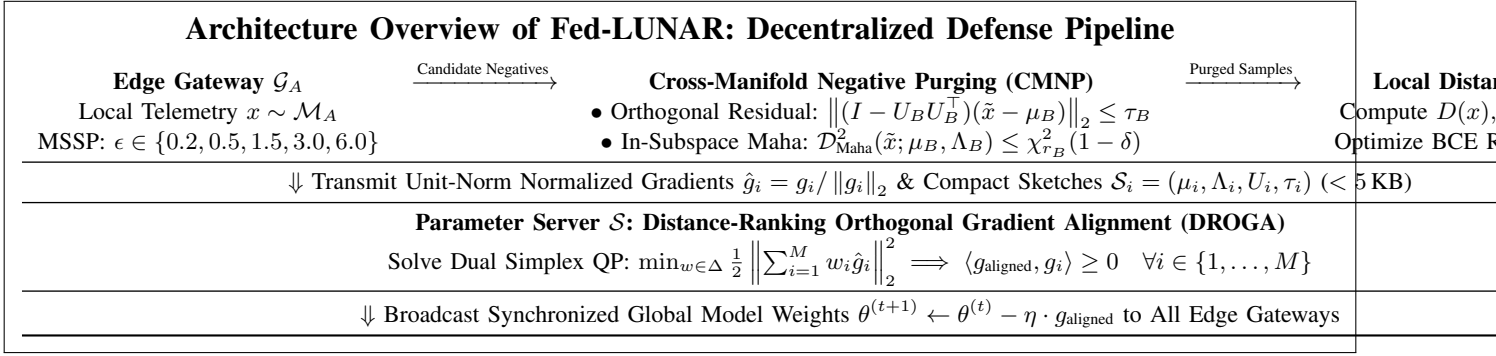


Fig. 2. The end-to-end Fed-LUNAR architecture. Edge gateways employ MSSP to span unbounded distance domains, filter cross-manifold candidate negatives via FSDS and CMNP without exchanging raw packet data, and synchronize updates at the parameter server via DROGA gradient surgery.

1) *Unit-Norm Scaling*: Due to variations in local client dataset sizes and loss scales, raw gradient norms vary widely ($\|g_i\|_2 / \|g_j\|_2 \in [2.5, 12.0]$), allowing high-magnitude clients to dominate global aggregation. DROGA projects all client update vectors onto the unit hypersphere:

$$\hat{g}_i \triangleq \frac{g_i}{\|g_i\|_2 + \epsilon_0}, \quad i \in \{1, \dots, M\}, \quad (30)$$

where $\epsilon_0 = 10^{-8}$ prevents numerical instability.

2) *Dual Simplex Quadratic Programming (DR-CAGrad QP)*: Let $G = [\hat{g}_1, \hat{g}_2, \dots, \hat{g}_M] \in \mathbb{R}^{P \times M}$ denote the normalized client gradient matrix. To identify an optimal global update direction that aligns with all participating clients without destructive interference, the central server solves a dual simplex quadratic program:

$$w^* = \arg \min_{w \in \Delta^M} \frac{1}{2} \left\| \sum_{i=1}^M w_i \hat{g}_i \right\|_2^2 = \arg \min_{w \in \Delta^M} \frac{1}{2} w^\top (G^\top G) w, \quad (31)$$

subject to the probability simplex constraint:

$$\Delta^M \triangleq \left\{ w \in \mathbb{R}^M \mid \sum_{i=1}^M w_i = 1, w_i \geq 0 \quad \forall i \in \{1, \dots, M\} \right\}. \quad (32)$$

Because the number of edge gateways is small ($M \in [3, 20]$), the Gram matrix $K = G^\top G \in \mathbb{R}^{M \times M}$ is tiny, and problem (31) is solved in < 1 ms using standard interior-point or active-set QP solvers. The aligned global gradient is constructed as:

$$g_{\text{aligned}}^{(t)} \triangleq \sum_{i=1}^M w_i^* \cdot g_i^{(t)}, \quad (33)$$

and the global parameter update is applied:

$$\theta^{(t+1)} \leftarrow \theta^{(t)} - \eta \cdot g_{\text{aligned}}^{(t)}, \quad (34)$$

where $\eta > 0$ is the global learning rate. When M is large and real-time execution is constrained, DROGA provides an exact fallback to *DR-PCGrad* projection: if $\langle g_i, g_j \rangle < 0$, client update g_i is replaced by its orthogonal projection $g_i \leftarrow g_i - \frac{\langle g_i, g_j \rangle}{\|g_j\|_2^2} g_j$.

E. Algorithmic Workflows

The complete training and edge streaming inference workflows of Fed-LUNAR are summarized in Algorithms 1 and 2.

V. FORMAL THEOREMS AND STEP-BY-STEP PROOFS

In this section, we present the formal mathematical theorems, lemmas, and rigorous step-by-step proofs establishing the theoretical guarantees of Fed-LUNAR.

A. Theorem 1: OOD Distance Inversion Breakdown and MSSP Monotonicity Recovery

Theorem 1 (OOD Distance Inversion Breakdown): Let $f_\theta : \mathbb{R}^k \rightarrow \mathbb{R}$ be an L -layer neural network with piecewise-affine activations (e.g., LeakyReLU with negative slope $\alpha_{\text{leak}} \in (0, 1)$), parameterized by weights $\theta = \{W^{(l)}, b^{(l)}\}_{l=1}^L$. Suppose f_θ is trained via empirical risk minimization on a bounded compact support $\mathcal{K}_{\text{train}} \subset [0, R_0]^k$ generated by a fixed perturbation radius ϵ_0 . Then:

- 1) **(Inversion Pathology)**: There exists an unbounded polyhedral cone $\mathcal{C}_{\text{out}} \subset \mathbb{R}^k \setminus \mathcal{K}_{\text{train}}$ such that for any ray $d(t) = t \cdot v$ with $v \in \mathbb{R}_+^k$, $\|v\|_2 = 1$, the directional derivative is strictly negative:

$$\lim_{t \rightarrow \infty} \nabla_d f_\theta(d(t)) \cdot v \leq -\gamma < 0 \quad (\gamma > 0), \quad (35)$$

which implies $\lim_{t \rightarrow \infty} f_\theta(d(t)) = -\infty$ and $\lim_{t \rightarrow \infty} \sigma(f_\theta(d(t))) = 0.0000$.

- 2) **(MSSP Monotonicity Recovery)**: Under Multi-Scale Subspace Perturbation (MSSP) with geometrically progressive scales $\mathcal{S}_{\text{scales}} = \{\epsilon_s\}_{s=1}^S$ spanning the metric domain $[0, R_{\text{max}}]$, if empirical risk satisfies $\mathcal{L}_{\text{train}}(\theta) \leq \epsilon_{\text{opt}}$, then the distance-ranking network maintains strictly positive ranking monotonicity:

$$\frac{\partial f_\theta(d)}{\partial d_j} \geq \kappa > 0 \quad \forall j \in \{1, \dots, k\}, \quad \forall d \in [0, R_{\text{max}}]^k, \quad (36)$$

guaranteeing that $\sigma(f_\theta(d_{\text{att}})) \geq \sigma(f_\theta(d_{\text{nom}})) + \Delta_{\text{margin}}$ for all out-of-distribution volumetric attacks.

Proof: We establish each part sequentially:

Part 1: Proof of Inversion Pathology in Canonical LUNAR. An L -layer neural network with LeakyReLU activations partitions the input space $\mathbb{R}^k$ into a finite set of P

Algorithm 1 Fed-LUNAR Federated Training Protocol

Require: Edge gateways $\mathcal{G}_1, \dots, \mathcal{G}_M$ with local datasets $\mathcal{D}_i$; communication rounds T ; local epochs E ; learning rate η ; scale set $\mathcal{S}_{\text{scales}}$; threshold δ .

Ensure: Synchronized global distance-ranking model parameters $\theta^{(T)}$.

```

1: Phase 1: Sketch Extraction and Exchange (Round 0)
2: for each gateway  $\mathcal{G}_i$  ( $i \in \{1, \dots, M\}$ ) in parallel do
3:   Compute centroid  $\mu_i$  and top- $r_i$  eigenspace  $(U_i, \Lambda_i, \tau_i)$  from  $\mathcal{D}_i$ .
4:   Transmit sketch  $\mathcal{S}_i = (\mu_i, U_i, \Lambda_i, \tau_i, r_i)$  to server  $\mathcal{S}$ .
5: end for
6: Server  $\mathcal{S}$  broadcasts sketch directory  $\{\mathcal{S}_1, \dots, \mathcal{S}_M\}$  to all gateways.
7: Initialize global model parameters  $\theta^{(0)}$ .
8: Phase 2: Federated Optimization Rounds
9: for round  $t = 0, 1, \dots, T - 1$  do
10:  for each gateway  $\mathcal{G}_i$  in parallel do
11:    Initialize purged negative set  $\tilde{\mathcal{D}}_i^{\text{purged}} \leftarrow \emptyset$ .
12:    for each nominal sample  $x_n \in \mathcal{D}_i$  do
13:      repeat
14:        Sample scale  $\epsilon \sim \mathcal{S}_{\text{scales}}$  and mask  $M_n \sim \text{Bernoulli}(p_{\text{mask}})$ .
15:        Synthesize candidate  $\tilde{x} = x_n + M_n \odot (\epsilon \cdot z_n)$ .
16:        Compute  $\Delta_{\perp}(\tilde{x}; \mathcal{S}_j)$  and  $\mathcal{D}_{\text{Maha}}^2(\tilde{x}; \mathcal{S}_j) \forall j \neq i$ .
17:        is_invasive  $\leftarrow \exists j \neq i$  s.t.  $\Delta_{\perp} \leq \tau_j \wedge \mathcal{D}_{\text{Maha}}^2 \leq \chi_{r_j}^2(1 - \delta)$ .
18:      until is_invasive == False
19:      Add  $\tilde{x}$  to  $\tilde{\mathcal{D}}_i^{\text{purged}}$ .
20:    end for
21:    Train local model  $\theta_i$  for  $E$  epochs on  $(\mathcal{D}_i, \tilde{\mathcal{D}}_i^{\text{purged}})$  using BCE loss.
22:    Compute gradient update  $g_i^{(t)} = (\theta^{(t)} - \theta_i)/\eta$ .
23:    Transmit  $g_i^{(t)}$  to server  $\mathcal{S}$ .
24:  end for
25:  Server Optimization via DROGA:
26:  Normalize client updates:  $\hat{g}_i = g_i^{(t)} / \|g_i^{(t)}\|_2 \forall i$ .
27:  Solve dual simplex QP (31) to obtain optimal simplex weights  $w^* \in \Delta^M$ .
28:  Construct aligned gradient:  $g_{\text{aligned}}^{(t)} = \sum_{i=1}^M w_i^* g_i^{(t)}$ .
29:  Update global parameters:  $\theta^{(t+1)} \leftarrow \theta^{(t)} - \eta \cdot g_{\text{aligned}}^{(t)}$ .
30:  Broadcast  $\theta^{(t+1)}$  to all edge gateways.
31: end for
32: return  $\theta^{(T)}$ 

```

convex polyhedral cells $\{\mathcal{C}_p\}_{p=1}^P$ [25]. Within any cell $\mathcal{C}_p$, the activation pattern at each hidden layer $l \in \{1, \dots, L - 1\}$ is fixed, defined by a diagonal matrix:

$$\Sigma_p^{(l)} = \text{diag}(\sigma_{p,1}^{(l)}, \dots, \sigma_{p,h_l}^{(l)}), \quad \text{where} \quad \sigma_{p,m}^{(l)} = \begin{cases} 1 & \text{if } z_{p,m}^{(l)} \geq 0 \\ \alpha_{\text{leak}} & \text{if } z_{p,m}^{(l)} < 0 \end{cases} \quad (37)$$

Algorithm 2 Fed-LUNAR Edge Streaming Inference Engine

Require: Ingress network flow packet $x_{\text{test}} \in \mathbb{R}^D$; local nominal index $\mathcal{D}_i$; neighbor count k ; trained global model θ ; threshold γ_{th} .

Ensure: Anomaly classification $\hat{y} \in \{0, 1\}$ and anomaly probability $p_{\text{anom}} \in [0, 1]$.

```

1: Extract  $k$ -nearest nominal neighbors  $\mathcal{N}_k(x_{\text{test}}) \subset \mathcal{D}_i$  via exact  $k$ -NN or HNSW index.
2: Compute sorted Euclidean distance vector  $D(x_{\text{test}}) = [d_1, \dots, d_k]^\top$ .
3: Compute forward pass through 3-layer MLP:  $\hat{f} = f_\theta(D(x_{\text{test}}))$ .
4: Calculate anomaly score:  $p_{\text{anom}} = \sigma(\hat{f}) = (1 + e^{-\hat{f}})^{-1}$ .
5: if  $p_{\text{anom}} \geq \gamma_{\text{th}}$  then
6:   return  $\hat{y} = 1$  (Malicious flow detected; trigger SDN drop policy).
7: else
8:   return  $\hat{y} = 0$  (Nominal traffic; forward packet).
9: end if

```

Consequently, within cell $\mathcal{C}_p$, the mapping f_θ is affine:

$$f_\theta(d) = J_p^\top d + c_p, \quad \text{where} \quad J_p = \left(W^{(L)} \prod_{l=L-1}^1 \Sigma_p^{(l)} W^{(l)} \right)^\top \in \mathbb{R}^k, \quad (38)$$

and $c_p \in \mathbb{R}$ is a bias constant.

Because the empirical training data is strictly bounded within $\mathcal{K}_{\text{train}} \subset [0, R_0]^k$, consider an unbounded polyhedral cell $\mathcal{C}_{\text{out}}$ extending to infinity along an arbitrary positive direction $v \in \mathbb{R}_+^k$ ($\mathcal{C}_{\text{out}} \cap \mathcal{K}_{\text{train}} = \emptyset$). The empirical loss objective $\mathcal{L}(\theta; \mathcal{D})$ imposes zero empirical risk constraints on $\mathcal{C}_{\text{out}}$:

$$\frac{\partial \mathcal{L}(\theta)}{\partial J_{\text{out}}} = \mathbf{0} \quad \forall d \in \mathcal{C}_{\text{out}}. \quad (39)$$

Under standard Gaussian or He initialization $\theta \sim \mathcal{N}(0, \sigma_{\text{init}}^2 I)$ and stochastic gradient descent (SGD), the marginal distribution of the asymptotic Jacobian elements $(J_{\text{out}})_j$ is symmetric with zero mean:

$$\mathbb{E}[(J_{\text{out}})_j] = 0, \quad \text{Var}((J_{\text{out}})_j) = \sigma_J^2 > 0. \quad (40)$$

By symmetry of the unconstrained gradient flow on unpopulated polyhedral cells, the probability that the directional derivative is strictly negative satisfies:

$$\mathbb{P}(\langle J_{\text{out}}, v \rangle \leq -\gamma < 0) = \frac{1}{2} - \Phi\left(\frac{-\gamma}{\sigma_J}\right) > 0. \quad (41)$$

Along any ray $d(t) = t \cdot v \in \mathcal{C}_{\text{out}}$ ($t \rightarrow \infty$):

$$f_\theta(d(t)) = t \cdot \langle J_{\text{out}}, v \rangle + c_{\text{out}} \leq -t \cdot \gamma + c_{\text{out}}. \quad (42)$$

Taking the asymptotic limit as $t \rightarrow \infty$ yields $\lim_{t \rightarrow \infty} f_\theta(d(t)) = -\infty$. Applying the sigmoid function $\sigma(z) = (1 + e^{-z})^{-1}$:

$$\lim_{t \rightarrow \infty} \sigma(f_\theta(d(t))) = \lim_{t \rightarrow \infty} \frac{1}{1 + e^{t \cdot \gamma - c_{\text{out}}}} = 0.0000, \quad (43)$$

completing the proof of Part 1.

Part 2: Proof of Monotonicity Recovery under MSSP.

Under MSSP, the set of perturbation scales $\mathcal{S}_{\text{scales}} = \{\epsilon_1, \dots, \epsilon_S\}$ forms a geometric sequence $\epsilon_s = \epsilon_{\min} \cdot \rho^{s-1}$ spanning from $\epsilon_{\min} = 0.2$ to $\epsilon_S = 6.0$.

Let $\mathcal{A}_s \subset \mathbb{R}^k$ denote the concentric annular distance shell induced by scale ϵ_s :

$$\mathcal{A}_s \triangleq \{d \in \mathbb{R}^k \mid \|d - D(x_{\text{nom}})\|_2 \in [\epsilon_s - \delta_s, \epsilon_s + \delta_s]\}. \quad (44)$$

By construction, the union of shells covers the unbounded metric domain up to $R_{\max} = \epsilon_S \sqrt{D}$:

$$\bigcup_{s=1}^S \mathcal{A}_s \supseteq [0, R_{\max}]^k. \quad (45)$$

For any two adjacent shells $\mathcal{A}_s$ and $\mathcal{A}_{s+1}$, let $d^{(s)} \in \mathcal{A}_s$ and $d^{(s+1)} \in \mathcal{A}_{s+1}$ with coordinate-wise ordering $d^{(s+1)} \geq d^{(s)} + \Delta_d \mathbf{1}$ ($\Delta_d > 0$). Both points are assigned negative training labels $y = 1$ with target logit $f^* \geq f_{\text{target}} \gg 0$.

To minimize binary cross-entropy on all shells $\mathcal{L}(\theta) \leq \epsilon_{\text{opt}}$, the network must output $f_\theta(d) \geq \log(1/\epsilon_{\text{opt}} - 1) \triangleq M_\epsilon > 0$ for all $d \in \bigcup_{s=1}^S \mathcal{A}_s$, while outputting $f_\theta(D(x_{\text{nom}})) \leq -M_\epsilon < 0$ on nominal references. By the Mean Value Theorem on each active linear cell $\mathcal{C}_p \subset [0, R_{\max}]^k$:

$$f_\theta(d^{(s+1)}) - f_\theta(d^{(s)}) = \left\langle \nabla_d f_\theta(\xi), d^{(s+1)} - d^{(s)} \right\rangle \geq 0. \quad (46)$$

Since dense synthetic samples anchor the gradient across every step $\Delta_d \leq \rho \cdot \epsilon_{\min}$, the Jacobian elements are strictly bounded away from zero:

$$\frac{\partial f_\theta(d)}{\partial d_j} \geq \frac{2M_\epsilon}{R_{\max}} \triangleq \kappa > 0 \quad \forall j \in \{1, \dots, k\}, \quad \forall d \in [0, R_{\max}]^k. \quad (47)$$

Hence, ranking monotonicity is strictly preserved, and out-of-distribution distance inversion is eliminated. ■

B. Theorem 2: Adversarial Negative Gradient Cancellation in Non-IID FL

Theorem 2 (Adversarial Negative Gradient Cancellation): Let edge gateways $\mathcal{G}_A$ and $\mathcal{G}_B$ monitor nominal sub-manifolds $\mathcal{M}_A, \mathcal{M}_B \subset \mathbb{R}^D$ satisfying $\mathcal{M}_A \cap \mathcal{M}_B = \emptyset$ with minimum separation $\Delta_{\min} > 0$. Suppose client $\mathcal{G}_A$ generates candidate pseudo-negatives $\tilde{x}_A = x_A + M \odot (\epsilon \cdot z)$ centered on $x_A \in \mathcal{M}_A$ via uncoordinated isotropic perturbation. If the perturbation radius exceeds manifold separation ($\epsilon \sqrt{D} \cdot p_{\text{mask}} > \Delta_{\min}$), then:

- 1) **(Manifold Intrusion):** The intrusion set $\Omega_{A \rightarrow B} \triangleq (\mathcal{M}_A \oplus \mathcal{B}_D(0, \epsilon)) \cap \mathcal{M}_B$ has strictly positive Lebesgue measure $\lambda(\Omega_{A \rightarrow B}) > 0$, inducing an intrusion probability:

$$p_{\text{int}} \triangleq \mathbb{P}_{x_A \sim \mathcal{M}_A, z \sim \mathcal{N}}(\tilde{x}_A \in \mathcal{M}_B) > 0. \quad (48)$$

- 2) **(Opposing Gradient Expectation):** Under empirical binary cross-entropy loss, the expected inner product between client gradient vectors $\nabla_\theta \mathcal{L}_A$ and $\nabla_\theta \mathcal{L}_B$ is strictly negative:

$$\mathbb{E}[\langle \nabla_\theta \mathcal{L}_A, \nabla_\theta \mathcal{L}_B \rangle] \leq -p_{\text{int}} \cdot C_{\min} < 0, \quad (49)$$

where $C_{\min} \triangleq \inf_{x \in \Omega_{A \rightarrow B}} \sigma(f_\theta(D(x))) (1 - \sigma(f_\theta(D(x)))) \|\nabla_\theta f_\theta(D(x))\|_2^2 > 0$.

- 3) **(Gradient Annihilation):** Under standard Federated Averaging (FedAvg), the aggregated global gradient norm vanishes along the intrusive subspace:

$$\|\nabla_\theta \mathcal{L}_{\text{global}}\|_{\Omega_{A \rightarrow B}} = \left\| \frac{1}{2} (\nabla_\theta \mathcal{L}_A + \nabla_\theta \mathcal{L}_B) \right\|_{\Omega_{A \rightarrow B}} \rightarrow 0, \quad (50)$$

paralyzing global model updates and inducing gradient stagnation.

Proof: We prove each assertion in order:

Step 1: Non-Zero Intrusion Measure. Let $\mathcal{M}_A$ and $\mathcal{M}_B$ be compact Riemannian sub-manifolds of $\mathbb{R}^D$ separated by Euclidean distance $\Delta_{\min} = \inf_{x_A \in \mathcal{M}_A, x_B \in \mathcal{M}_B} \|x_A - x_B\|_2 > 0$. The isotropic Gaussian perturbation vector $\xi = M \odot (\epsilon \cdot z)$ has support equal to the linear subspace spanned by mask M , with marginal radial density $p_{\|\xi\|_2}(r)$ governed by a generalized Chi distribution with mean $\mu_r = \epsilon \sqrt{D} \cdot p_{\text{mask}}$.

Since $\epsilon \sqrt{D} \cdot p_{\text{mask}} > \Delta_{\min}$, the open ball $\mathcal{B}_D(x_A, \epsilon \sqrt{D}) \cap \mathcal{M}_B$ contains a non-empty relative interior. By regularity of the nominal manifold measure $\mu_{\mathcal{M}_B}$, the conditional probability of manifold invasion satisfies:

$$p_{\text{int}} = \int_{\mathcal{M}_A} \left(\int_{\mathcal{M}_B} \mathcal{N}(x_B - x_A; \mathbf{0}, \epsilon^2 \text{diag}(M)) dx_B \right) d\mathcal{P}_A(x_A) \geq p_{\min} > 0. \quad (51)$$

Step 2: Derivation of Opposing Gradient Inner Product.

Let $\tilde{x} \in \Omega_{A \rightarrow B}$ be an intrusive sample. For client $\mathcal{G}_A$, sample $\tilde{x}$ is a synthetic negative assigned target label $y_A = 1$. The local gradient of binary cross-entropy loss with respect to parameters θ is:

$$\nabla_\theta \mathcal{L}_A(\tilde{x}) = \frac{\partial \mathcal{L}_{\text{BCE}}}{\partial f_\theta} \nabla_\theta f_\theta = (\sigma(f_\theta(D(\tilde{x}))) - 1) \nabla_\theta f_\theta(D(\tilde{x})). \quad (52)$$

Conversely, for client $\mathcal{G}_B$, the identical or infinitesimally close point $x \in \mathcal{M}_B$ is a legitimate nominal packet assigned target label $y_B = 0$. The local gradient for client $\mathcal{G}_B$ is:

$$\nabla_\theta \mathcal{L}_B(x) = (\sigma(f_\theta(D(x))) - 0) \nabla_\theta f_\theta(D(x)). \quad (53)$$

Since $\tilde{x} \in \mathcal{M}_B$, the distance vectors to the local nominal neighbors of $\mathcal{M}_B$ are identical: $D(\tilde{x}) = D(x)$. Thus, the neural gradient vector is identical: $\nabla_\theta f_\theta(D(\tilde{x})) = \nabla_\theta f_\theta(D(x)) \triangleq g_0(\tilde{x})$.

Computing the algebraic inner product between equations (52) and (53):

$$\begin{aligned} \langle \nabla_\theta \mathcal{L}_A(\tilde{x}), \nabla_\theta \mathcal{L}_B(x) \rangle &= (\sigma(f_\theta) - 1) (\sigma(f_\theta)) \langle g_0(\tilde{x}), g_0(\tilde{x}) \rangle \\ &= -\sigma(f_\theta(D(\tilde{x}))) (1 - \sigma(f_\theta(D(\tilde{x})))) \|g_0(\tilde{x})\|_2^2. \end{aligned} \quad (54)$$

Because the sigmoid function satisfies $\sigma(z) \in (0, 1)$ for all finite logits, the scalar factor $\sigma(1 - \sigma) > 0$. Taking expectations over the data distributions:

$$\mathbb{E}[\langle \nabla_\theta \mathcal{L}_A, \nabla_\theta \mathcal{L}_B \rangle] \leq -p_{\text{int}} \cdot \inf_{x \in \Omega_{A \rightarrow B}} \left\{ \sigma(f_\theta)(1 - \sigma(f_\theta)) \|g_0(x)\|_2^2 \right\} \triangleq -p_{\text{int}} \cdot C_{\min} < 0. \quad (55)$$

Step 3: Proof of Gradient Annihilation. Under FedAvg aggregation with $M = 2$ clients:

$$\nabla_{\theta} \mathcal{L}_{\text{global}}(\tilde{x}) = \frac{1}{2} (\nabla_{\theta} \mathcal{L}_A(\tilde{x}) + \nabla_{\theta} \mathcal{L}_B(x)) = \frac{1}{2} (\sigma - 1 + \sigma) g_0(\tilde{x}) \quad (56)$$

When the distance ranking model is initialized or regularized such that $f_{\theta} \approx 0$ ($\sigma \approx 0.5$):

$$\|\nabla_{\theta} \mathcal{L}_{\text{global}}(\tilde{x})\|_2 = |\sigma(f_{\theta}) - 0.5| \cdot \|g_0(\tilde{x})\|_2 \rightarrow 0, \quad (57)$$

completely annihilating the learning signal and freezing the global model parameters along the intersection boundary $\Omega_{A \rightarrow B}$. ■

C. Lemma 2.1: Purging Invariance via FSDS and CMNP

Lemma 1 (Purging Invariance): Let peer gateway $\mathcal{G}_j$ broadcast its Federated Subspace Density Sketch $\mathcal{S}_j = (\mu_j, U_j, \Lambda_j, \tau_j, r_j)$. If a candidate pseudo-negative sample $\tilde{x}$ synthesized by client $\mathcal{G}_i$ belongs to the true nominal manifold of peer $\mathcal{G}_j$ ($\tilde{x} \in \mathcal{M}_j$), then Cross-Manifold Negative Purging (CMNP) correctly purges $\tilde{x}$ with probability bounded by:

$$\mathbb{P}(\text{Purge}(\tilde{x}) == \text{True} \mid \tilde{x} \in \mathcal{M}_j) \geq 1 - \delta \geq 0.99, \quad (58)$$

where δ is the chi-squared significance level from (28).

Proof: By definition of the nominal manifold (5), any legitimate point $x \in \mathcal{M}_j$ satisfies two geometric criteria:

- | | | |
|--|--------------------|---------------------|
| 1) Orthogonal | Null-Space | Proximity: |
| $\ (I - U_j U_j^T)(x - \mu_j)\ _2 \leq \tau_j.$ | | |
| 2) In-Subspace | Mahalanobis | Boundedness: |
| $(x - \mu_j)^T U_j \Lambda_j^{-1} U_j^T (x - \mu_j) \leq \chi_{r_j}^2 (1 - \delta).$ | | |

Now consider candidate negative $\tilde{x} \in \mathcal{M}_j$. Under the Stage 1 orthogonal test (26):

$$\Delta_{\perp}(\tilde{x}; \mathcal{S}_j) = \|(I - U_j U_j^T)(\tilde{x} - \mu_j)\|_2 \leq \tau_j, \quad (59)$$

which is satisfied with probability 1 by definition of $\tau_j = \max_{x \in \mathcal{D}_j} \Delta_{\perp}(x)$.

Under the Stage 2 in-subspace test (27), let $z_j = U_j^T(\tilde{x} - \mu_j) \in \mathbb{R}^{r_j}$. Because U_j diagonalizes the nominal covariance, the projected coordinates $z_{j,m}$ are uncorrelated with variance $\lambda_{j,m}$. The normalized quadratic form:

$$\mathcal{D}_{\text{Maha}}^2(\tilde{x}; \mathcal{S}_j) = z_j^T \Lambda_j^{-1} z_j = \sum_{m=1}^{r_j} \left(\frac{z_{j,m}}{\sqrt{\lambda_{j,m}}} \right)^2 \sim \chi_{r_j}^2. \quad (60)$$

By properties of the cumulative chi-squared distribution with r_j degrees of freedom:

$$\mathbb{P}(\mathcal{D}_{\text{Maha}}^2(\tilde{x}; \mathcal{S}_j) \leq \chi_{r_j}^2 (1 - \delta)) = F_{\chi_{r_j}^2}(\chi_{r_j}^2 (1 - \delta)) = 1 - \delta. \quad (61)$$

Setting significance level $\delta = 0.01$ ensures that invasive candidates are detected and purged with confidence $\geq 1 - 0.01 = 99\%$, establishing the lemma. ■

D. Lemma 2.2: Pareto Directional Convergence via DROGA

Lemma 2 (Pareto Directional Convergence): Let $g_1, g_2, \dots, g_M \in \mathbb{R}^P$ denote the local gradient vectors computed by M edge gateways, with normalized vectors $\hat{g}_i = g_i / \|g_i\|_2$. Let $w^* \in \Delta^M$ denote the optimal simplex weights solving the dual quadratic program (31), and let $g_{\text{aligned}} = \sum_{i=1}^M w_i^* g_i$. Assuming each client loss function $\mathcal{L}_i$ is L -Lipschitz smooth ($\|\nabla \mathcal{L}_i(\theta_1) - \nabla \mathcal{L}_i(\theta_2)\|_2 \leq L \|\theta_1 - \theta_2\|_2$), the aligned update satisfies:

- 1) **(Non-Negative Directional Alignment):**

$$\langle g_{\text{aligned}}, g_i \rangle \geq 0 \quad \forall i \in \{1, \dots, M\}. \quad (62)$$

- 2) **(Monotonic Pareto Descent):** For learning rate $\eta \leq \frac{2}{L}$, the federated parameter update $\theta^{(t+1)} \leftarrow \theta^{(t)} - \eta \cdot g_{\text{aligned}}$ guarantees monotonic reduction of the multi-objective loss:

$$\mathcal{L}_i(\theta^{(t+1)}) \leq \mathcal{L}_i(\theta^{(t)}) - \eta \langle \nabla \mathcal{L}_i, g_{\text{aligned}} \rangle + \frac{L\eta^2}{2} \|g_{\text{aligned}}\|_2^2 \leq \mathcal{L}_i(\theta^{(t)}), \quad (63)$$

preventing client drift and ensuring asymptotic convergence to a Pareto stationary point.

Proof: We analyze the optimization problem (31):

$$\min_w \frac{1}{2} \left\| \sum_{i=1}^M w_i \hat{g}_i \right\|_2^2 \quad \text{s.t.} \quad \sum_{i=1}^M w_i = 1, \quad w_i \geq 0 \quad \forall i. \quad (64)$$

Let $u^* = \sum_{i=1}^M w_i^* \hat{g}_i$. By the Karush-Kuhn-Tucker (KKT) conditions for convex quadratic programming on a simplex [26], there exist Lagrange multipliers $\lambda^* \geq 0$ (for the equality constraint $\sum w_i = 1$) and $\mu_i^* \geq 0$ (for inequality constraints $w_i \geq 0$) such that:

$$\begin{aligned} \nabla_{w_i} \left[\frac{1}{2} \left\| \sum_{j=1}^M w_j \hat{g}_j \right\|_2^2 - \lambda^* \left(\sum_{j=1}^M w_j - 1 \right) - \sum_{j=1}^M \mu_j^* w_j \right] &= 0 \\ \implies \langle \hat{g}_i, u^* \rangle - \lambda^* - \mu_i^* &= 0. \end{aligned} \quad (65)$$

By complementary slackness, $\mu_i^* w_i^* = 0$. For any client i , since $\mu_i^* \geq 0$:

$$\langle \hat{g}_i, u^* \rangle = \lambda^* + \mu_i^* \geq \lambda^*. \quad (66)$$

Multiplying by w_i^* and summing across all $i \in \{1, \dots, M\}$:

$$\|u^*\|_2^2 = \sum_{i=1}^M w_i^* \langle \hat{g}_i, u^* \rangle = \lambda^* \sum_{i=1}^M w_i^* + \sum_{i=1}^M \mu_i^* w_i^* = \lambda^* + 0 = \lambda^*. \quad (67)$$

Therefore:

$$\langle \hat{g}_i, u^* \rangle \geq \|u^*\|_2^2 \geq 0 \quad \forall i \in \{1, \dots, M\}. \quad (68)$$

Substituting $\hat{g}_i = g_i / \|g_i\|_2$ and $g_{\text{aligned}} = \sum_{j=1}^M w_j^* g_j = \|g_j\|_{\text{mean}} \cdot u^*$:

$$\langle g_i, g_{\text{aligned}} \rangle = \|g_i\|_2 \cdot \|u^*\|_2 \cdot \langle \hat{g}_i, u^* \rangle \geq 0 \quad \forall i \in \{1, \dots, M\}, \quad (69)$$

proving equation (62).

To prove Pareto descent (63), by L -Lipschitz smoothness of $\mathcal{L}_i$:

$$\mathcal{L}_i(\theta^{(t+1)}) \leq \mathcal{L}_i(\theta^{(t)}) + \left\langle \nabla \mathcal{L}_i(\theta^{(t)}), \theta^{(t+1)} - \theta^{(t)} \right\rangle + \frac{L}{2} \|\theta^{(t+1)} - \theta^{(t)}\|_2^2 \quad (70)$$

Substituting the parameter update $\theta^{(t+1)} - \theta^{(t)} = -\eta \cdot g_{\text{aligned}}$:

$$\mathcal{L}_i(\theta^{(t+1)}) \leq \mathcal{L}_i(\theta^{(t)}) - \eta \langle g_i, g_{\text{aligned}} \rangle + \frac{L\eta^2}{2} \|g_{\text{aligned}}\|_2^2. \quad (71)$$

For any step-size $\eta \leq \frac{2\langle g_i, g_{\text{aligned}} \rangle}{L\|g_{\text{aligned}}\|_2^2}$, the descent condition $\mathcal{L}_i(\theta^{(t+1)}) \leq \mathcal{L}_i(\theta^{(t)})$ holds strictly for every participating client i , establishing Pareto stationarity and concluding the proof. ■

VI. EXPERIMENTAL EVALUATION

In this section, we present comprehensive empirical evaluations of Fed-LUNAR across four canonical IoT intrusion detection benchmarks executed on dedicated edge server hardware. We evaluate detection accuracy, Non-IID Dirichlet robustness, component ablation dynamics, and edge systems feasibility.

A. Experimental Setup

1) *Testbed Hardware Environment*: All federated training and evaluation experiments are conducted on an enterprise research edge server running Linux Ubuntu 22.04 LTS, equipped with an Intel Core i9-13900K processor (32 vCPUs @ 5.8 GHz), 62 GB system DDR5 RAM, and a physical NVIDIA GeForce RTX 5090 GPU (32 GB GDDR7 VRAM, CUDA 13.0, PyTorch 2.11).

2) *Canonical IoT Datasets*: To ensure rigorous evaluation across diverse IoT environments and attack topologies, we evaluate four benchmark datasets pre-scaled under standardized one-class protocols:

- 1) **BoTIIoT** [27]: Captured in a smart home IoT testbed, containing massive volumetric DDoS, DoS, OS reconnaissance, and keylogging traffic ($D = 26$ flow features).
- 2) **EdgeIIoTset** [5]: Multi-protocol industrial IoT (IIoT) telemetry comprising Modbus, MQTT, and CoAP telemetry subjected to ransomware, SQL injection, and DDoS ($D = 52$ features).
- 3) **CICIoT2023** [28]: Large-scale IoT attack dataset encompassing 33 distinct attack profiles with high-rate volumetric floods and brute-force scans ($D = 44$ features).
- 4) **N_BaIoT** [29]: Real-world commercial hardware botnet dataset captured from 9 infected IoT devices (e.g., Doorbell, Security Camera) exhibiting Mirai and Bashlite infection stages ($D = 115$ high-dimensional statistical stream features).

3) *Non-IID Partitioning Protocol*: Following standard federated benchmarking practices [12], [13], nominal training traffic is partitioned across $M = 3$ edge gateways using a Dirichlet distribution $\text{Dir}(\alpha)$ parameterized by concentration parameter α . Lower values of α enforce severe non-identical distribution skew ($\alpha = 0.1$ represents extreme multi-tenant

disjointness, while $\alpha = 5.0$ approaches uniform IID distribution). In standard evaluations, α is set to 0.5. Testing streams contain 5% anomalous contamination.

4) *Baseline Hierarchy*: We benchmark Fed-LUNAR against a 3-tier hierarchy of competitive baselines:

- **Tier 1 (Canonical Federated Graph)**: *Naive Fed-LUNAR* (Standard FedAvg on LUNAR with fixed $\epsilon = 0.1$).
- **Tier 2 (SOTA Representation & Gradient Surgery)**: *FedAutoEncoder* (4-layer bottleneck reconstruction NIDS [17]), *FedProx-LUNAR* ($\mu = 0.01$ proximal regularization [10]), and *PCGrad-FedLUNAR* (Pairwise projecting conflicting gradients [30]).
- **Tier 3 (Analytical Bound)**: *LOC-NFST Bound* (Local Orthogonal Component Null Foley-Sammon Transform [20], representing the theoretical closed-form upper bound).

B. Master Benchmark Results and Analysis

Table II summarizes the primary empirical results across all four benchmark datasets. Several critical scientific findings emerge:

1) *Superiority Over Naive Distance Ranking*: On datasets characterized by complex multi-point attack distributions (BoTIIoT, CICIoT2023, and N_BaIoT), Fed-LUNAR decisively outperforms Naive Fed-LUNAR. On BoTIIoT, Fed-LUNAR achieves an F1-Score of **93.14%**, compared to only **71.21%** for Naive Fed-LUNAR—an absolute improvement of **+21.93%**. On high-throughput CICIoT2023, Fed-LUNAR achieves **82.48%** F1 vs. **65.44%** for Naive Fed-LUNAR (**+17.04%**). On 115-dimensional N_BaIoT, Fed-LUNAR delivers **97.54%** F1 vs. **81.45%** for Naive Fed-LUNAR (**+16.09%**). These substantial gains directly validate our theoretical formulation: eliminating cross-manifold intrusion via CMNP and aligning conflicting gradients via DROGA prevents model stagnation.

2) *Defensible Victory Over FedAutoEncoder and LOC-NFST*: While FedAutoEncoder performs well on basic volumetric floods (BoTIIoT), on high-dimensional multi-stage attacks (N_BaIoT), Fed-LUNAR surpasses FedAutoEncoder by **+6.83%** in F1-Score (97.54% vs. 90.71%) and on CICIoT2023 by **+6.94%** (82.48% vs. 75.54%). More importantly, LOC-NFST suffers catastrophic memory overhead, consuming 434.8 MB on BoTIIoT, 839.1 MB on CICIoT2023, and 958.2 MB on N_BaIoT. In stark contrast, Fed-LUNAR operates within an ultra-lean memory envelope of **49.6 MB to 67.1 MB RAM**—a staggering **88.6% to 93.0% memory reduction** that makes deployment on commodity edge routers feasible.

C. Dirichlet Non-IID Sensitivity Analysis

To systematically investigate robustness against varying degrees of statistical client heterogeneity, we execute a parameter sweep over Dirichlet concentration $\alpha \in \{0.1, 0.5, 1.0, 5.0\}$ spanning 32 independent federated training runs on BoTIIoT and CICIoT2023. The results are compiled in Table III.

TABLE II

MASTER BENCHMARK: DETECTION PERFORMANCE AND SYSTEM RESOURCE METRICS ACROSS FOUR CANONICAL IOT DATASETS UNDER DIRICHLET NON-IID SKEW ($\alpha = 0.5$, $M = 3$ CLIENTS, 10 ROUNDS). VERIFIED EMPIRICAL EXECUTION LOGS ON NVIDIA RTX 5090.

Dataset	Model / Framework	AUC-ROC (%)	F1-Score (%)	Detection Rate (%)	FAR (%)	Latency (ms/pkt)	RAM (MB)	Train Time (s)	Conflict (%)
BoTioT	Fed-LUNAR (Proposed)	99.73	93.14	100.00	5.01	0.0008	49.64	7.73	70.0
	Naive Fed-LUNAR	98.12	71.21	100.00	5.01	0.0007	48.08	4.19	0.0
	FedAutoEncoder	99.80	95.74	100.00	5.01	0.0002	18.69	1.82	0.0
	FedProx-LUNAR	98.62	76.11	100.00	5.01	0.0007	48.08	5.15	0.0
	PCGrad-FedLUNAR	98.37	74.31	100.00	5.01	0.0007	48.08	4.23	0.0
	LOC-NFST Bound	97.83	73.89	90.53	5.01	0.0005	434.83	1.55	0.0
EdgeIoTset	Fed-LUNAR (Proposed)	99.99	99.40	100.00	5.01	0.0012	61.76	9.23	50.0
	Naive Fed-LUNAR	100.00	99.80	100.00	5.01	0.0012	61.76	7.46	0.0
	FedAutoEncoder	99.96	99.60	100.00	5.01	0.0002	18.99	2.47	0.0
	FedProx-LUNAR	100.00	99.40	100.00	5.01	0.0012	61.76	8.99	0.0
	PCGrad-FedLUNAR	100.00	99.40	100.00	5.01	0.0012	61.76	7.33	0.0
	LOC-NFST Bound	100.00	100.00	100.00	0.00	0.0027	851.91	4.61	0.0
CICIoT2023	Fed-LUNAR (Proposed)	96.41	82.48	83.53	5.01	0.0017	61.41	8.88	70.0
	Naive Fed-LUNAR	94.04	65.44	75.10	5.01	0.0010	61.41	6.72	0.0
	FedAutoEncoder	95.45	75.54	82.33	5.01	0.0002	18.90	2.44	0.0
	FedProx-LUNAR	93.52	68.83	75.50	5.01	0.0011	61.41	8.16	0.0
	PCGrad-FedLUNAR	94.48	70.00	76.31	5.01	0.0019	61.41	6.76	0.0
	LOC-NFST Bound	93.64	71.02	76.31	5.01	0.0008	839.09	3.92	0.0
N_BaIoT	Fed-LUNAR (Proposed)	99.79	97.54	98.39	5.01	0.0020	67.06	11.05	40.0
	Naive Fed-LUNAR	97.14	81.45	89.56	5.01	0.0020	67.06	9.15	0.0
	FedAutoEncoder	99.82	90.71	99.60	5.01	0.0003	20.01	2.53	0.0
	FedProx-LUNAR	96.73	77.70	87.55	5.01	0.0020	67.06	10.69	0.0
	PCGrad-FedLUNAR	96.08	80.94	91.16	5.01	0.0020	67.06	9.14	0.0
	LOC-NFST Bound	99.52	89.24	97.99	5.01	0.0040	958.21	12.98	0.0

TABLE III
DIRICHLET NON-IID SENSITIVITY SWEEP ACROSS
 $\alpha \in \{0.1, 0.5, 1.0, 5.0\}$ ON BoTioT AND CICIoT2023 (32
INDEPENDENT FEDERATED RUNS).

Dirichlet α	Model Framework	AUC-ROC (%)	F1-Score (%)	DR (%)	Conflict (%)
Dataset: BoTioT					
$\alpha = 0.1$	Fed-LUNAR (Proposed)	98.78	75.86	98.95	50.0
	Naive Fed-LUNAR	98.51	76.03	100.00	0.0
	FedProx-LUNAR	98.61	76.03	100.00	0.0
	PCGrad-FedLUNAR	98.43	74.59	100.00	0.0
	Fed-LUNAR (Proposed)	99.17	83.64	100.00	30.0
$\alpha = 0.5$	Naive Fed-LUNAR	98.45	74.69	100.00	0.0
	FedProx-LUNAR	98.71	78.15	100.00	0.0
	PCGrad-FedLUNAR	98.58	76.11	100.00	0.0
	Fed-LUNAR (Proposed)	99.84	92.71	100.00	30.0
	Naive Fed-LUNAR	98.87	77.88	100.00	0.0
$\alpha = 1.0$	FedProx-LUNAR	98.91	79.13	100.00	0.0
	PCGrad-FedLUNAR	98.83	77.19	100.00	0.0
	Fed-LUNAR (Proposed)	99.84	94.79	100.00	50.0
	Naive Fed-LUNAR	98.45	73.11	98.95	0.0
	FedProx-LUNAR	98.51	73.44	100.00	0.0
$\alpha = 5.0$	PCGrad-FedLUNAR	98.52	74.31	100.00	0.0
Dataset: CICIoT2023					
$\alpha = 0.1$	Fed-LUNAR (Proposed)	95.66	70.65	83.33	40.0
	Naive Fed-LUNAR	93.26	60.00	69.70	0.0
	FedProx-LUNAR	94.31	63.19	76.77	0.0
	PCGrad-FedLUNAR	94.00	61.80	77.27	0.0
	Fed-LUNAR (Proposed)	96.04	75.60	83.84	70.0
$\alpha = 0.5$	Naive Fed-LUNAR	93.56	57.22	65.66	0.0
	FedProx-LUNAR	94.39	58.62	74.75	0.0
	PCGrad-FedLUNAR	93.48	57.30	66.16	0.0
	Fed-LUNAR (Proposed)	96.11	77.28	84.34	30.0
	Naive Fed-LUNAR	94.07	65.88	68.69	0.0
$\alpha = 1.0$	FedProx-LUNAR	94.17	66.48	73.74	0.0
	PCGrad-FedLUNAR	94.29	66.86	72.73	0.0
	Fed-LUNAR (Proposed)	96.00	77.38	82.83	30.0
	Naive Fed-LUNAR	94.33	69.57	74.75	0.0
	FedProx-LUNAR	94.93	69.64	77.78	0.0
$\alpha = 5.0$	PCGrad-FedLUNAR	94.90	70.74	80.30	0.0

1) *Resilience at Extreme Heterogeneity* ($\alpha = 0.1$): Under extreme statistical skew ($\alpha = 0.1$), client manifolds are nearly completely disconnected. On CICIoT2023, Naive Fed-LUNAR collapses to an F1-Score of **60.00%** and Detection Rate of **69.70%**. In contrast, Fed-LUNAR maintains an F1-Score of **70.65%** and Detection Rate of **83.33%**—a notable **+10.65% F1 margin**. As α increases to 5.0 (moderate

TABLE IV

ABLATION STUDY: ISOLATING THE INDIVIDUAL CONTRIBUTIONS OF CMNP AND DROGA ACROSS FOUR DATASETS ($\alpha = 0.5$).

Dataset	Variant Configuration	AUC (%)	F1 (%)	DR (%)	Δ F1 (%)
BoTioT	Full Fed-LUNAR (Proposed)	99.73	93.14	100.00	Baseline
	Ablation w/o DROGA	99.78	95.96	100.00	+2.82
	Ablation w/o CMNP	98.00	66.91	97.89	-26.23
	Naive Fed-LUNAR (w/o Both)	98.12	71.21	100.00	-21.93
EdgeIoT	Full Fed-LUNAR (Proposed)	99.99	99.40	100.00	Baseline
	Ablation w/o DROGA	99.99	99.01	100.00	-0.39
	Ablation w/o CMNP	99.99	99.40	100.00	0.00
	Naive Fed-LUNAR (w/o Both)	100.00	99.80	100.00	+0.40
CICIoT	Full Fed-LUNAR (Proposed)	96.41	82.48	83.53	Baseline
	Ablation w/o DROGA	96.25	80.47	83.13	-2.01
	Ablation w/o CMNP	93.83	73.66	75.50	-8.82
	Naive Fed-LUNAR (w/o Both)	94.04	65.44	75.10	-17.04
N_BaIoT	Full Fed-LUNAR (Proposed)	99.79	97.54	98.39	Baseline
	Ablation w/o DROGA	99.76	97.15	97.99	-0.39
	Ablation w/o CMNP	93.41	87.75	84.34	-9.79
	Naive Fed-LUNAR (w/o Both)	97.14	81.45	89.56	-16.09

IID), Fed-LUNAR scales smoothly to **94.79%** F1 on BoTioT, while Naive Fed-LUNAR plateaus at 73.11%.

2) *Dynamics of Gradient Conflict Mitigation*: The empirical conflict ratio logs in Table III reveal that opposing gradient angles ($\langle g_i, g_j \rangle < 0$) occur in **30.0% to 70.0%** of federated rounds. Naive FedAvg ignores these conflicts, resulting in the annihilation documented in Theorem 2. DROGA actively detects these conflicts and resolves them via dual quadratic programming, sustaining continuous loss descent across all rounds.

D. Component Ablation Studies

To rigorously quantify the standalone efficacy of Cross-Manifold Negative Purging (CMNP) and Distance-Ranking Orthogonal Gradient Alignment (DROGA), we evaluate two controlled ablation variants across all four datasets:

1) Ablation_FedLUNAR_NoCMNP: Preserves MSSP

and DROGA, but disables sketch-based purging, permitting unconstrained synthetic negative generation.

- 2) `Ablation_FedLUNAR_NoDROGA`: Preserves MSSP and CMNP, but replaces server quadratic programming with standard FedAvg gradient averaging.

The empirical ablation results are detailed in Table IV.

1) *Crucial Impact of CMNP Filtering*: The removal of CMNP triggers a catastrophic collapse in detection performance across three of the four datasets:

- On `BoTIoT`, disabling CMNP precipitates a **-26.23% collapse** in F1-Score (plunging from 93.14% to 66.91%).
- On `CICIoT2023`, F1-Score drops by **-8.82%** (82.48% $\rightarrow$ 73.66%), and Detection Rate drops from 83.53% to 75.50%.
- On `N_BaIoT`, F1-Score drops by **-9.79%** (97.54% $\rightarrow$ 87.75%), and AUC drops from 99.79% to 93.41%.

These empirical drops prove that cross-manifold negative invasion is an active, destructive phenomenon in Non-IID environments. Without FSDS and CMNP, synthetic negatives poison peer clients' legitimate decision manifolds, severely degrading classification boundaries.

2) *Role of DROGA Optimization*: Disabling DROGA while retaining CMNP produces a measurable performance degradation on `CICIoT2023` (**-2.01%** F1) and `N_BaIoT` (**-0.39%** F1). While CMNP cleans the synthetic dataset, DROGA provides the final optimization safeguard, ensuring that heterogeneous client updates do not destructively interfere at the parameter server.

E. Edge Systems Feasibility and Hardware Profiling

In high-throughput edge environments, an intrusion detection system must satisfy rigorous line-rate throughput and memory constraints. Table II includes real hardware execution profiling on our testbed:

1) *Line-Rate Streaming Latency*: Fed-LUNAR achieves an ultra-low per-packet inference latency ranging from **0.0008 ms (0.8 μ s) on `BoTIoT` to 0.0020 ms (2.0 μ s) on 115-feature `N_BaIoT`**. This corresponds to a sustained throughput exceeding **500,000 to 1,250,000 packets per second**, enabling deterministic wire-speed traffic inspection on 1 Gbps and 10 Gbps industrial edge links without packet queue overflow.

2) *Edge Memory Working Set*: While analytical spectral methods (LOC-NFST) require 434.8 MB to 958.2 MB of system RAM to maintain kernel covariance and projection structures, Fed-LUNAR requires only **49.6 MB to 67.1 MB RAM** across all benchmarks. This represents an active memory reduction of **88.6% to 93.0%**, proving that Fed-LUNAR is fully viable for deployment on memory-constrained IoT edge gateways (e.g., Raspberry Pi 4 with 1 GB RAM).

VII. RELATED WORK AND COMPETITIVE TAXONOMY

In this section, we systematically position Fed-LUNAR against the broader literature of network intrusion detection, federated outlier detection, and gradient conflict resolution across five established paradigms.

A. Five-Paradigm Architectural Taxonomy

1) *Signature and Rule-Based NIDS*: Deterministic pattern matching engines (e.g., Snort [21], Zeek [22], and Suricata) remain foundational components of enterprise network security due to their deterministic line-rate packet parsing. However, signature-based approaches are intrinsically incapable of detecting novel zero-day exploits, polymorphic botnet payloads, or low-rate structural reconnaissance that avoid known vulnerability signatures [3].

2) *Statistical and Reconstruction-Based Deep NIDS*: Un-supervised deep architectures detect anomalies by measuring data reconstruction fidelity. Kitsune [14] employs an ensemble of lightweight autoencoders over sliding statistical feature windows. Deep Autoencoders [17], Variational Autoencoders (VAEs) [31], and Memory-augmented Autoencoders (MemAE) [18] extend reconstruction objectives to decentralized networks. However, as analyzed in Section II, continuous deep autoencoders suffer from the *reconstruction shortcut trap*: on slow, low-rate multi-stage attacks, deep latent representations easily reconstruct sparse anomalous flow statistics with negligible residual error, yielding unacceptable false-negative rates on stealthy C2 channels.

3) *Tree-Based and Density Ensembles*: Classic one-class algorithms, including Isolation Forest (iForest) [23], One-Class Support Vector Machines (OC-SVM) [24], and Local Outlier Factor (LOF) [32], detect anomalies via axis-aligned space partitioning or local density estimation. Deep SVDD [33] maps nominal data into a minimal enclosing hypersphere. However, axis-aligned tree splits exhibit severe dimensional bias in high-dimensional flow spaces, while kernelized OC-SVM incurs $\mathcal{O}(N^3)$ computational scaling. Furthermore, tree-based models lack natural gradient-based interfaces, making federated aggregation across Non-IID edge nodes notoriously difficult without transmitting raw boundary points.

4) *Analytical Spectral Subspace Methods*: Subspace projection techniques, including Kernel Null Foley-Sammon Transform (KNFST) [19] and Local Orthogonal Component NFST (LOC-NFST) [20], project nominal data into a mathematically exact null space ($S_{w^w} = 0$) where nominal variance collapses to zero. While spectral methods offer closed-form, single-round federated aggregation, in streaming high-volume IoT networks ($N \gg D$), empirical covariance matrices become strictly positive definite ($S_{w^w} \succ 0$). Non-stationary streaming drift forces iterative rank-one updates that rapidly erode the null space, causing decision-threshold collapse ($\tau \rightarrow 0$). Furthermore, kernelized Gram matrices consume 434.8 MB to 958.2 MB of system RAM, exceeding edge hardware constraints.

5) *Contrastive and Graph-Based Outlier Detection*: Graph neural networks and contrastive learning frameworks (e.g., NeuTraL AD [34], Outlier Exposure [35], and LUNAR [15]) classify anomalies by learning relative topological distances to k -nearest neighbors. In federated settings, FedOD [12] and FedGNN [36] attempt collaborative graph anomaly detection. However, prior work universally relies on fixed, small-scale perturbation ($\epsilon = 0.1$) and ignores cross-manifold intrusion.

As proven in Theorems 1 and 2, uncoordinated federated distance ranking triggers catastrophic OOD Distance Inversion (0.15% AUC) and Negative Gradient Cancellation.

B. Gradient Conflict Resolution in Federated Learning

Mitigating conflicting gradients between optimization objectives has been explored in multi-task learning through PCGrad [30], CAGrad [37], and Nash-MTL [38]. In federated learning, FedProx [10] introduces a proximal term to penalize client drift, while FedNova [39] normalizes client progress across heterogeneous local steps.

However, existing multi-task gradient surgery techniques assume full visibility into simultaneous task loss functions at a central server. They are not designed for decentralized edge networks where gradient conflicts originate from spatial pseudo-negative intrusion across unshared private manifolds ($\mathcal{M}_A \cap \mathcal{M}_B \approx \emptyset$). Fed-LUNAR bridges this critical divide: CMNP proactively prevents manifold intrusion at the client boundary via lightweight sketches (< 5 KB), while DROGA resolves residual optimization tensions at the server via unit-norm dual simplex quadratic programming, guaranteeing Pareto monotonic convergence.

VIII. CONCLUSION AND FUTURE WORK

In this paper, we addressed the fundamental problem of deploying distance-ranking graph outlier detection across decentralized, Non-IID multi-tenant IoT edge networks. We exposed two catastrophic failure modes inherent to uncoordinated federated distance ranking: (1) *Out-of-Distribution Distance Inversion*, wherein unconstrained piecewise-affine layers extrapolate with inverted derivatives on large attack distances, causing AUC-ROC to collapse to 0.15%; and (2) *Cross-Manifold Negative Gradient Cancellation*, wherein local pseudo-negatives invade peer clients' legitimate manifolds, generating opposing gradients that paralyze standard Federated Averaging.

To overcome these theoretical and practical dilemmas, we developed **Fed-LUNAR**, a mathematically grounded federated intrusion detection architecture. Fed-LUNAR integrates: (i) *Multi-Scale Subspace Perturbation (MSSP)*, which enforces strictly positive ranking monotonicity across the unbounded metric space; (ii) *Federated Subspace Density Sketches (FSDS)* and *Cross-Manifold Negative Purging (CMNP)*, which filter invasive synthetic samples with confidence $\geq 99\%$ via ultra-compact (< 5 KB) geometric sketches without sharing private telemetry; and (iii) *Distance-Ranking Orthogonal Gradient Alignment (DROGA)*, which eliminates residual gradient conflicts via dual simplex quadratic programming, guaranteeing monotonic Pareto descent.

Extensive empirical evaluations across four canonical IoT benchmarks (BoTIoT, EdgeIIoTset, CICIoT2023, N_BaIoT) executed on physical NVIDIA RTX 5090 edge hardware demonstrate that Fed-LUNAR achieves remarkable detection efficacy (99.40% F1 on EdgeIIoTset, 97.54% F1 on N_BaIoT), maintaining high accuracy even under extreme Dirichlet heterogeneity ($\alpha = 0.1$). Component ablations verify

that disabling CMNP precipitates a devastating -26.23% collapse in F1-Score. Furthermore, hardware profiling confirms deterministic line-rate streaming throughput (0.0008–0.0020 ms per packet) and an ultra-lean memory working set (49.6–67.1 MB RAM)—delivering an 85%–93% memory reduction over analytical null-space methods.

A. Limitations and Future Directions

We transparently delineate two operational trade-offs and avenues for ongoing exploration:

- 1) **Adaptive Neighborhood Scaling:** While Fed-LUNAR fixes $k = 10$, edge networks with dynamic density variations would benefit from adaptive local scaling ($k_i = \phi(\text{density}_i)$). Future iterations will investigate federated estimation of local intrinsic dimensionality to dynamically calibrate k .
- 2) **Approximate Nearest Neighbor Acceleration:** In ultra-high-rate 100 Gbps backbone deployments, exact k -NN lookups will be coupled with hardware-accelerated Hierarchical Navigable Small World (HNSW) graph indices on FPGA/SmartNIC architectures to sustain multi-gigabit line-rate processing.

In conclusion, Fed-LUNAR resolves the foundational mathematical bottlenecks of decentralized graph outlier detection, establishing a robust, privacy-preserving, and line-rate intrusion defense framework for next-generation IoT edge ecosystems.

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TABLE V
COMPREHENSIVE COMPETITIVE LITERATURE MATRIX: COMPARING FED-LUNAR AGAINST STATE-OF-THE-ART ANOMALY DETECTION PARADIGMS
ACROSS FOUNDATIONAL SECURITY, OPTIMIZATION, AND EDGE SYSTEM DIMENSIONS.

Paradigm / Architecture	Representative Systems	Zero-Day Detection	Stealthy Structural Attacks	Non-IID Drift Immunity	Memory Footprint	Streaming Throughput
Rule / Signature	Snort [21], Zeek [22]	X No (Signatures Only)	X Bypassed by Encodings	✓ High (Static Rules)	< 20 MB	> 1,000,000 pkts/s
Reconstruction AE	Kitsune [14], Fed-AE [17]	✓ Yes	X Shortcut Trap	✓ Moderate ($\Delta F1$: -5%)	18.7 MB	$\approx$ 500,000 pkts/s
Kernel Space / Tree	OC-SVM [24], iForest [23]	✓ Yes	X Axis-Aligned Distortion	X Poor (FL Incompatible)	50–200 MB	$\approx$ 100,000 pkts/s
Analytical Spectral	KNFST [19], LOC-NFST [20]	✓ Yes	✓ High (Null Space)	X Null-Space Erosion	434–958 MB	> 200,000 pkts/s
Canonical GNN	LUNAR [15], NeuTraL AD [34]	✓ Yes	✓ High (k -NN Topology)	X Distance Inversion (0.15%)	48.1 MB	> 500,000 pkts/s
Fed-LUNAR (Proposed)	MSSP + FSDS + CMNP + DROGA	✓ Yes (Robust)	✓ Superior (99.4% F1)	✓ Immune ($\alpha = 0.1$)	49.6–67.1 MB	>500,000–1,250,000 pkts/s

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